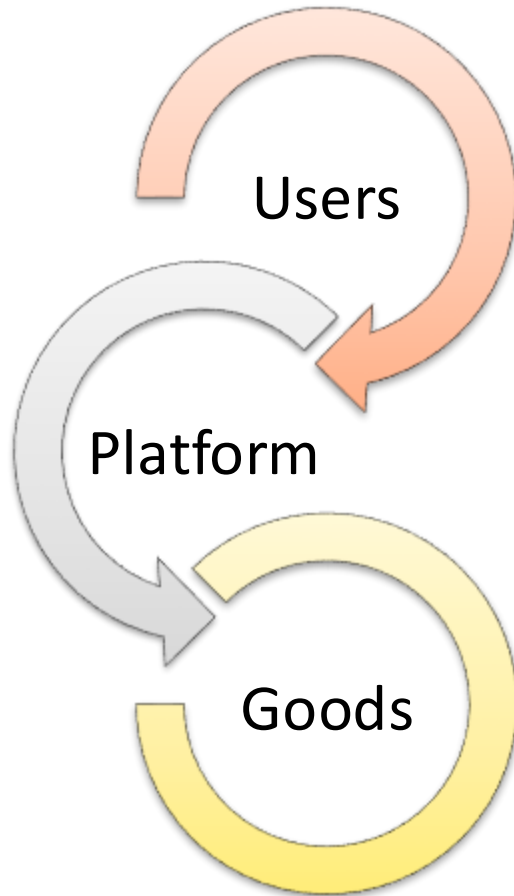




WIESŁAW PIETRUSZKIEWICZ

MAIN ASPECTS OF DATA-DRIVEN E-COMMERCE

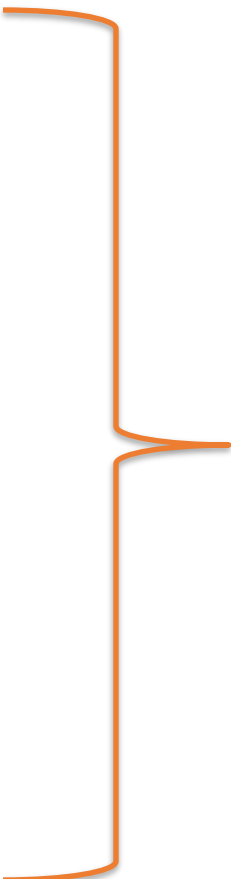
Who – How – What



- The success of e-commerce depends on proper interaction of 3 elements
- Orientation towards data gives ability to improve this interaction
- Therefore, we aim at answering – who and how does what?

Improvements

- Customer understanding
- Range of products and services matching demands
- Efficient interaction
- Cross-selling (suggested, not enforced)
- Up-selling (tempting upgrades)

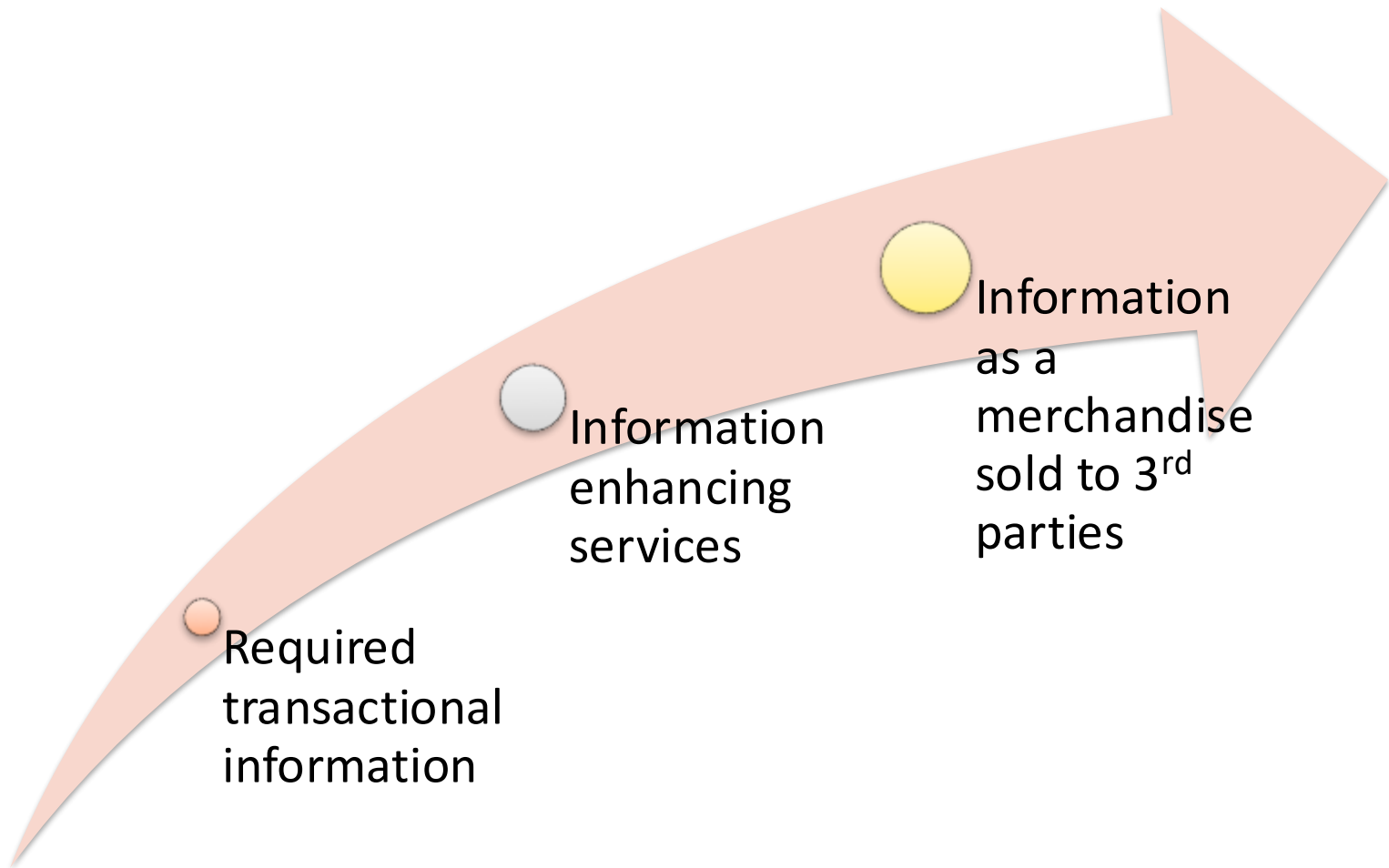


More sales

Notes – In this presentation ...

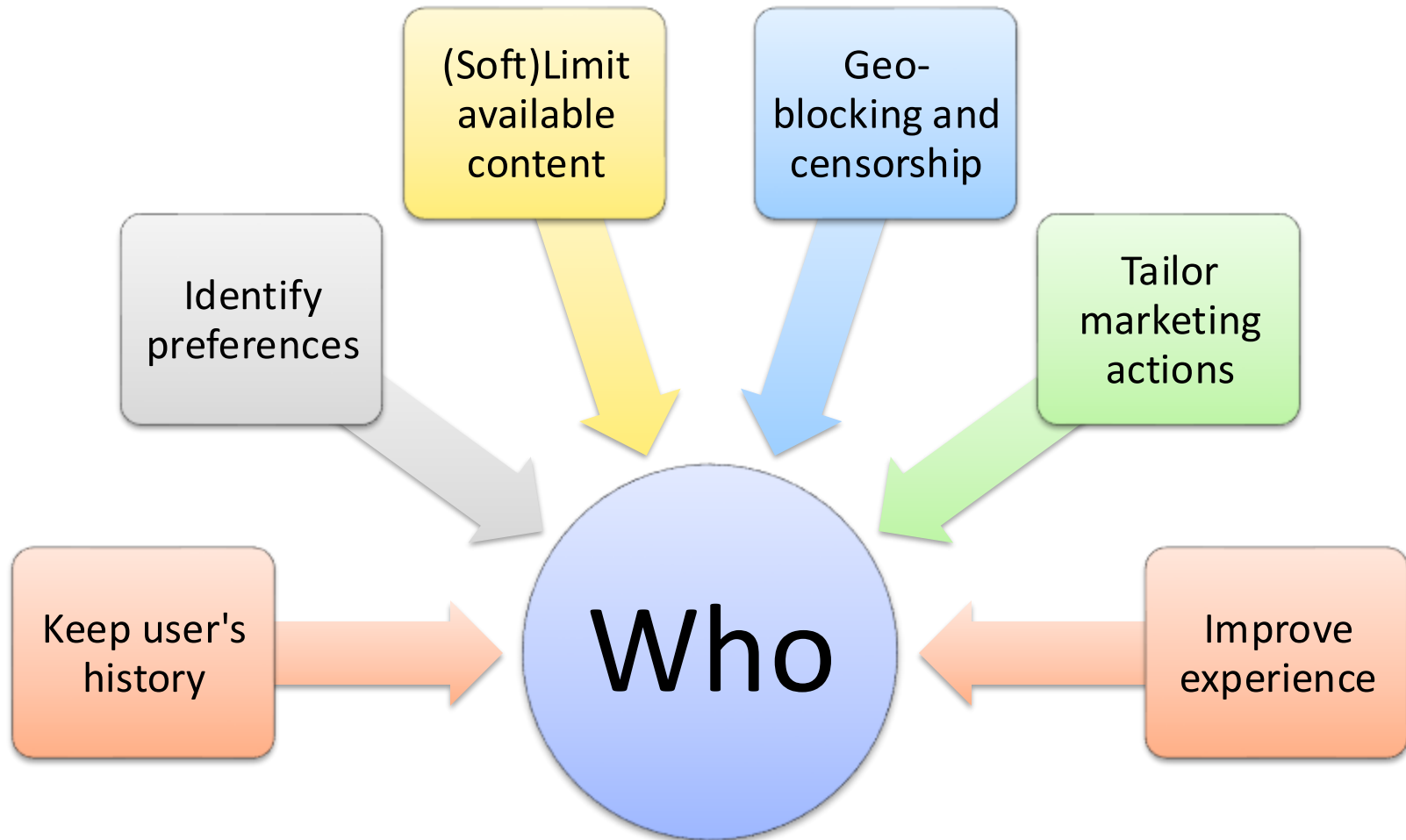
- The terms of items, products, services or goods are interchangeable
- The same applies to users, customers and clients
- A wide range of possible data-driven solutions is covered. However, their usage depends on many individual factors (technological and business ones)

Types of information



USERS ANALYSIS

Reasons for “Who”

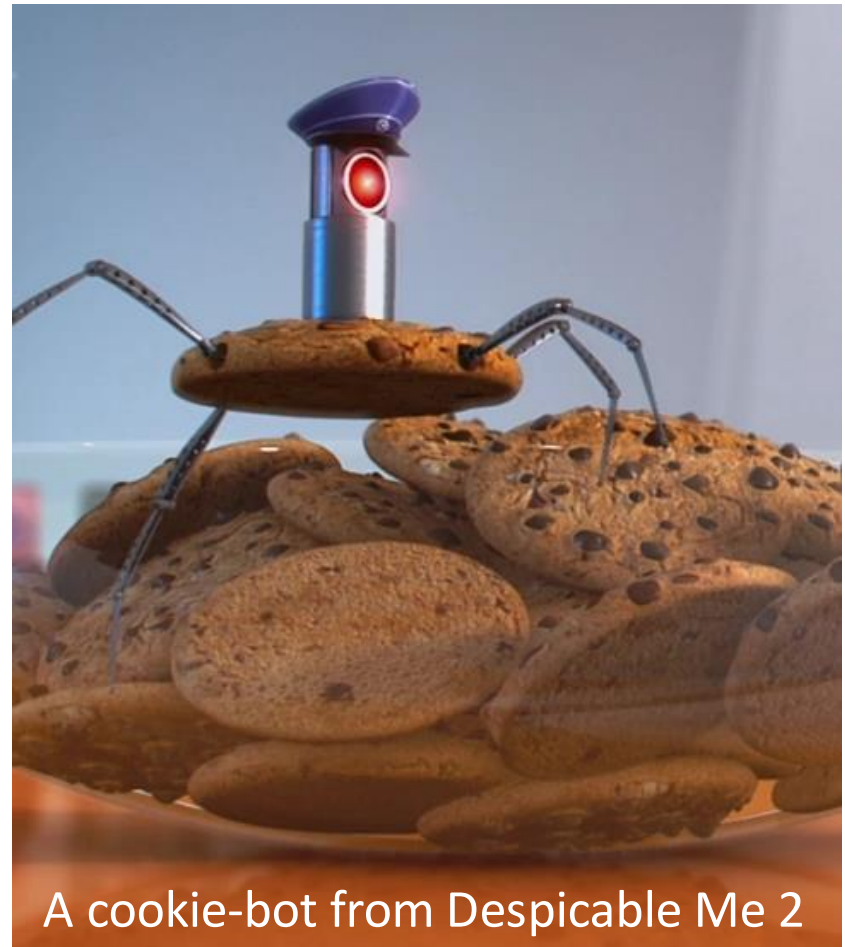


Known vs Unknown User

- Running webservices, we want to track users, and adjust the content/functionality
- Naïve solution – enforce registration & logging
- How to track non-logged users?
- Identified machine $\neq$ identified user i.e.:
 - 1 user can use more than 1 machine
 - 1 machine can have more than 1 user
- Issues with shared accounts “*Netflix case*”

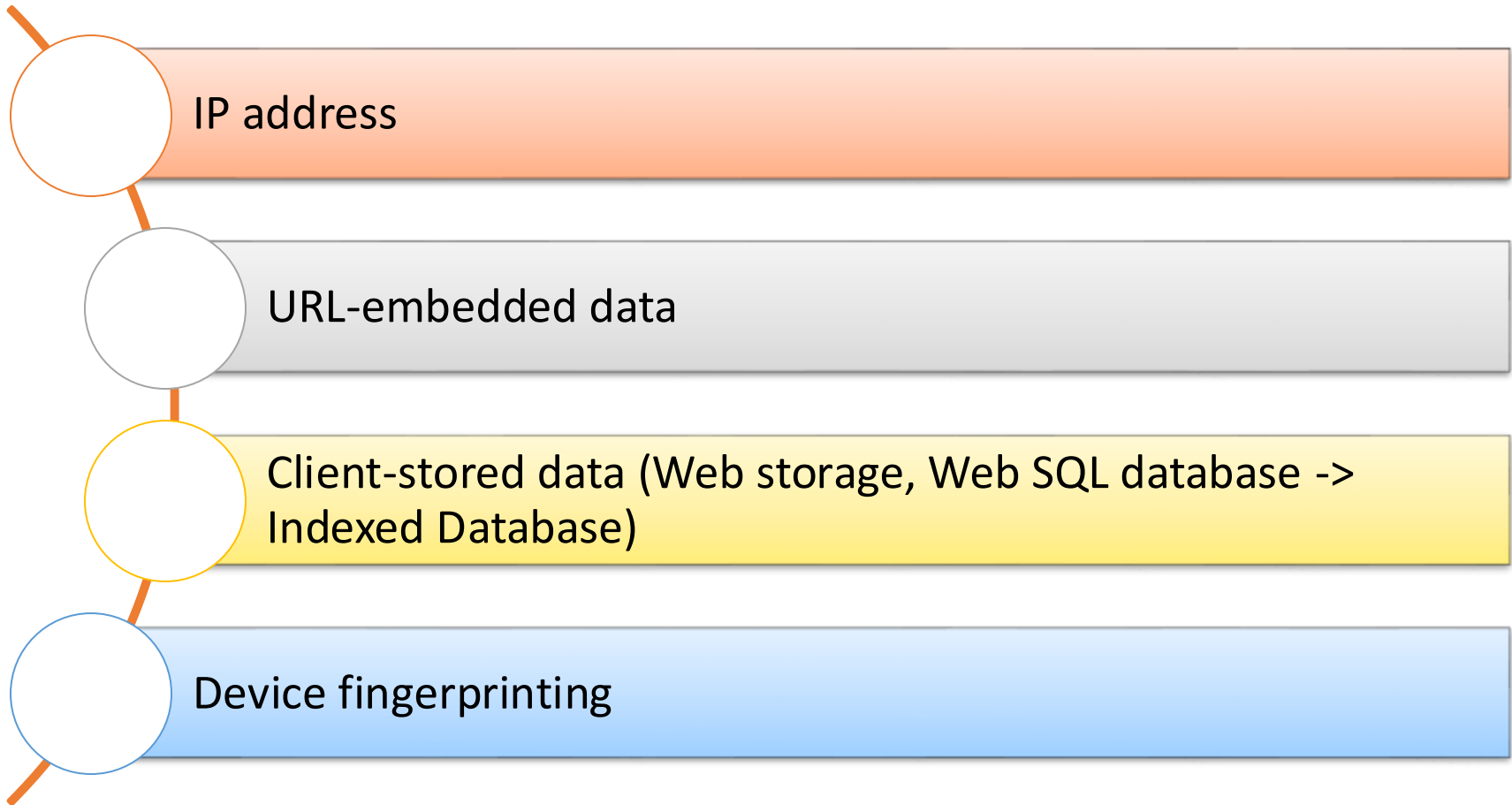
HTTP Cookies

- The most classic approach to tracking
- Introduced in 1997
- Small textual information stored by and retrieved from browser
- Accessible only to a source domain or its ancestors



A cookie-bot from Despicable Me 2

Alternatives to Cookies

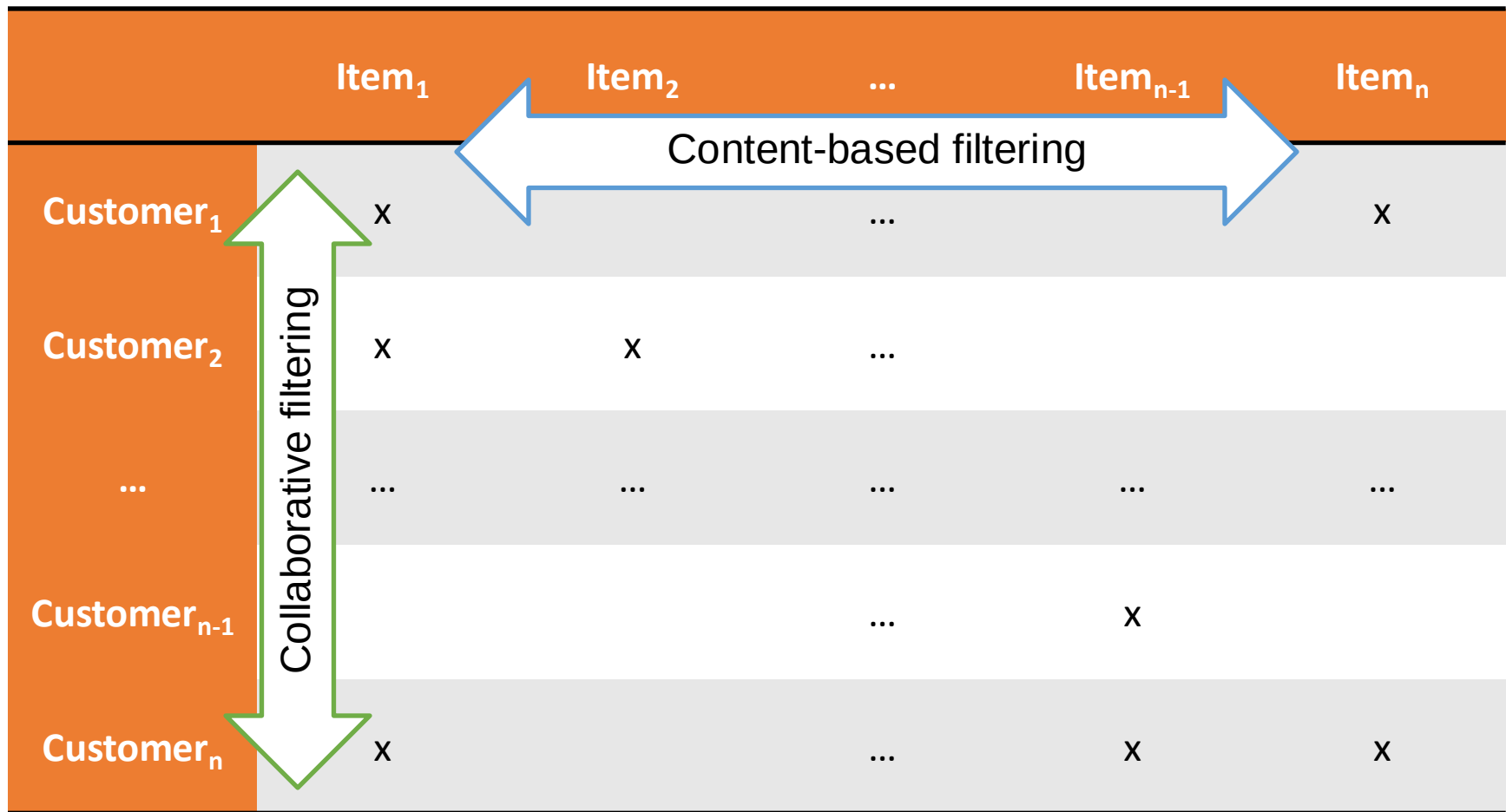


CONTENT - RECOMMENDATIONS

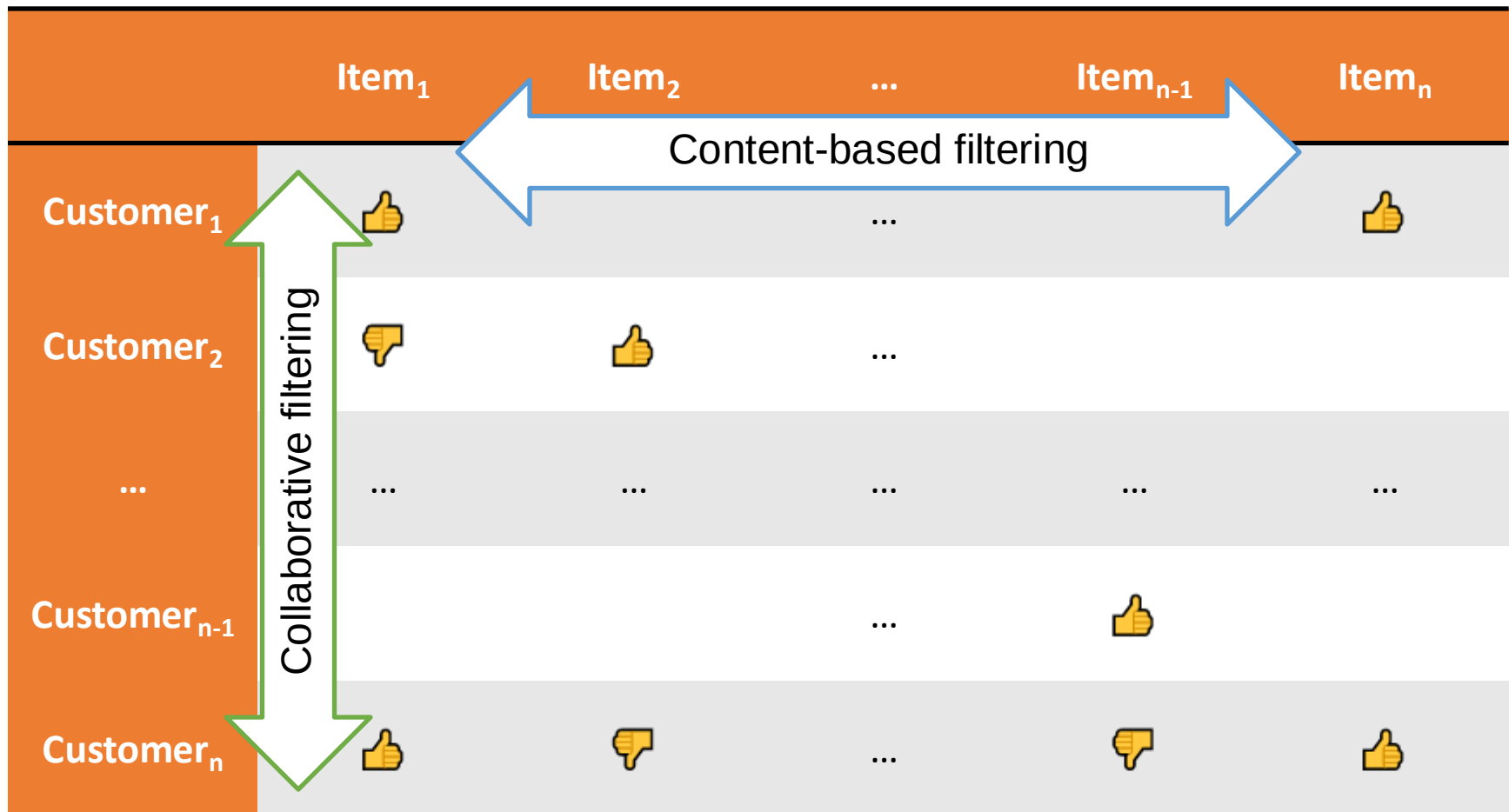
Recommender/recommendation Systems

- Recommender system is a system predicting preference/rating of an item given by users
- A variety of application in e-commerce
- The most common one – recommendation of products in web-shops
- In this scenario – recommender matches two sides i.e. users with items
- ~~Non personalised RSes – Typical Top10s~~

Customers vs Items – Relationship



Customers vs Items – Ranked Relationship



RS – Main Approaches

Collaborative filtering

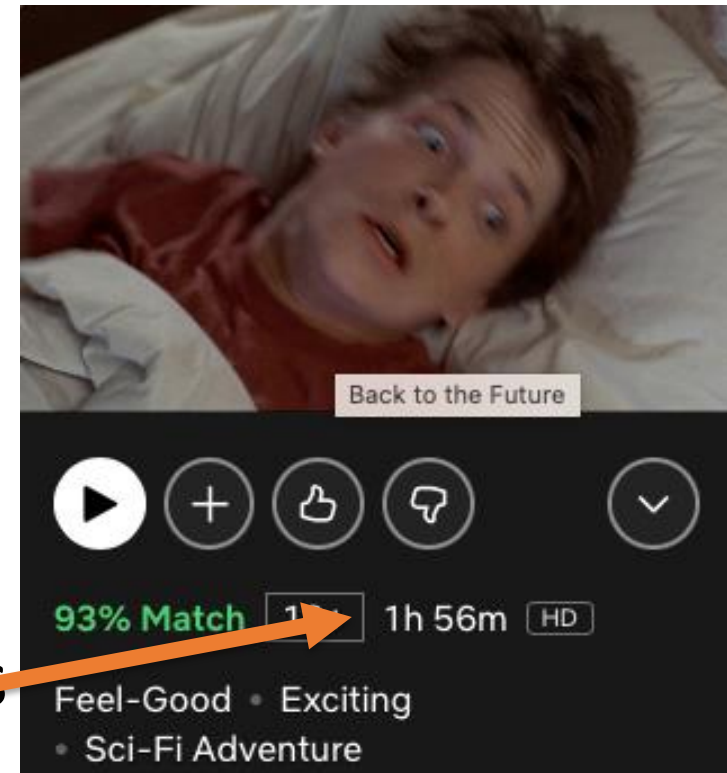
- Build models using users' past behaviour
- Data foundation:
 - Past items selected by user
 - Activity of other users
- Having ratings recommender searches for similar users
- Predicts rating for items
- Or an algorithm can be item-centric

Content filtering

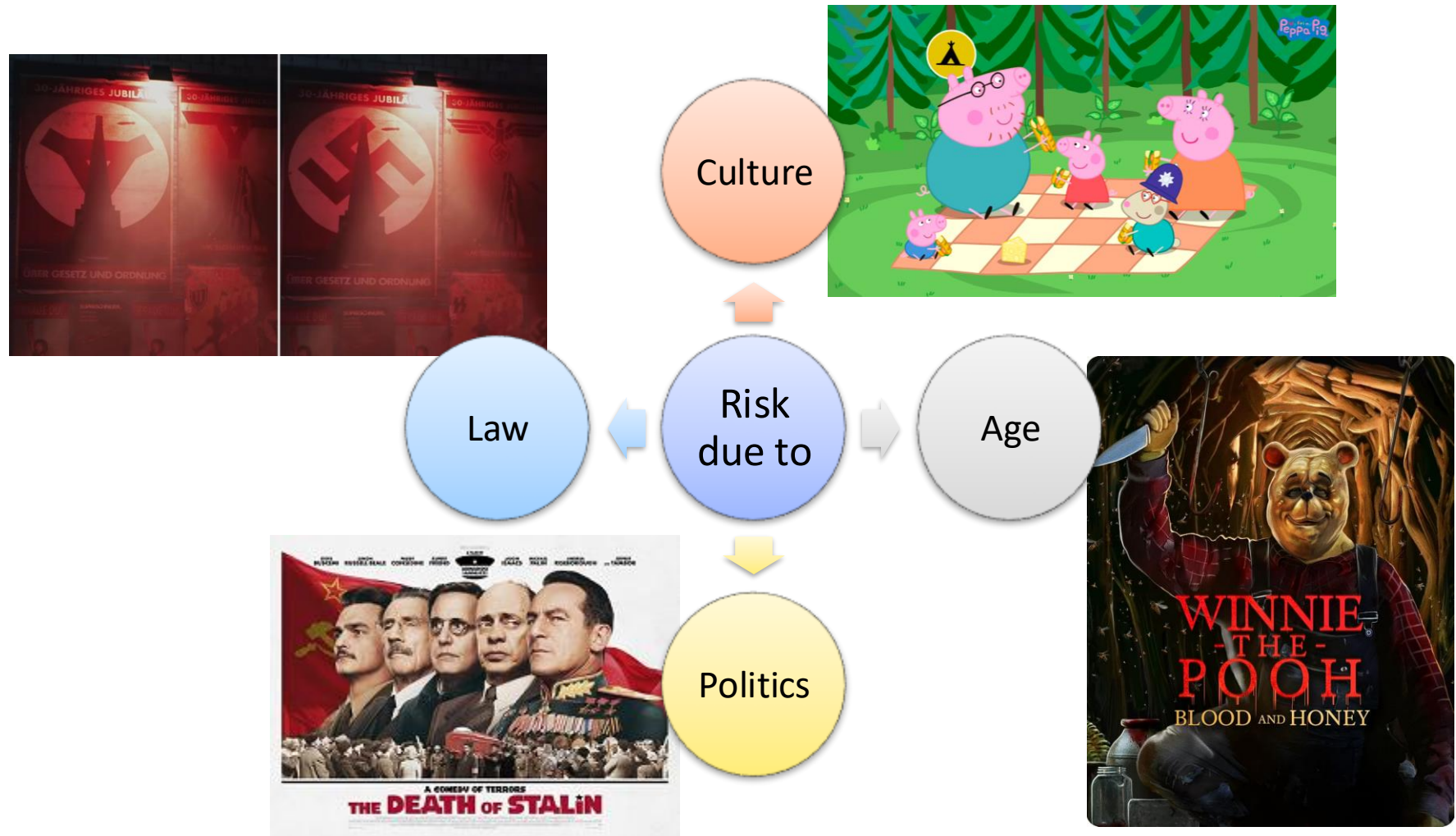
- Items are described with a set of features
- Knowing what user selects algorithm creates user's profile
- Selected items are closely related to previously selected ones
- **+ AI-based approaches (a broader topic)**

RS CF –Ideas

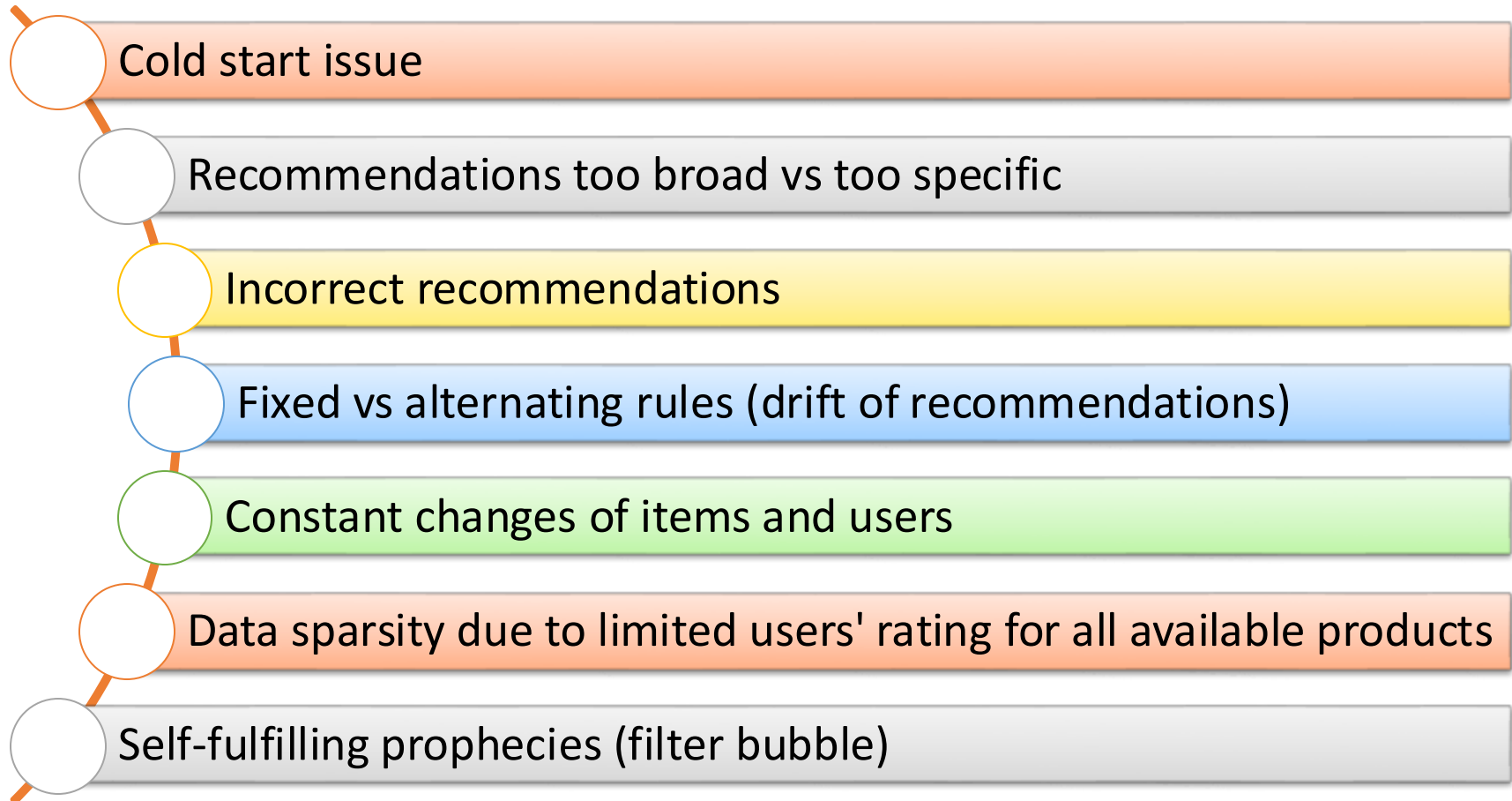
- Data collection:
 - explicit (requested)
 - implicit (observed)
- Data sources:
 - single source (service)
 - multiple sources (services)
- A re-evaluation done by users (reinforcement learning)



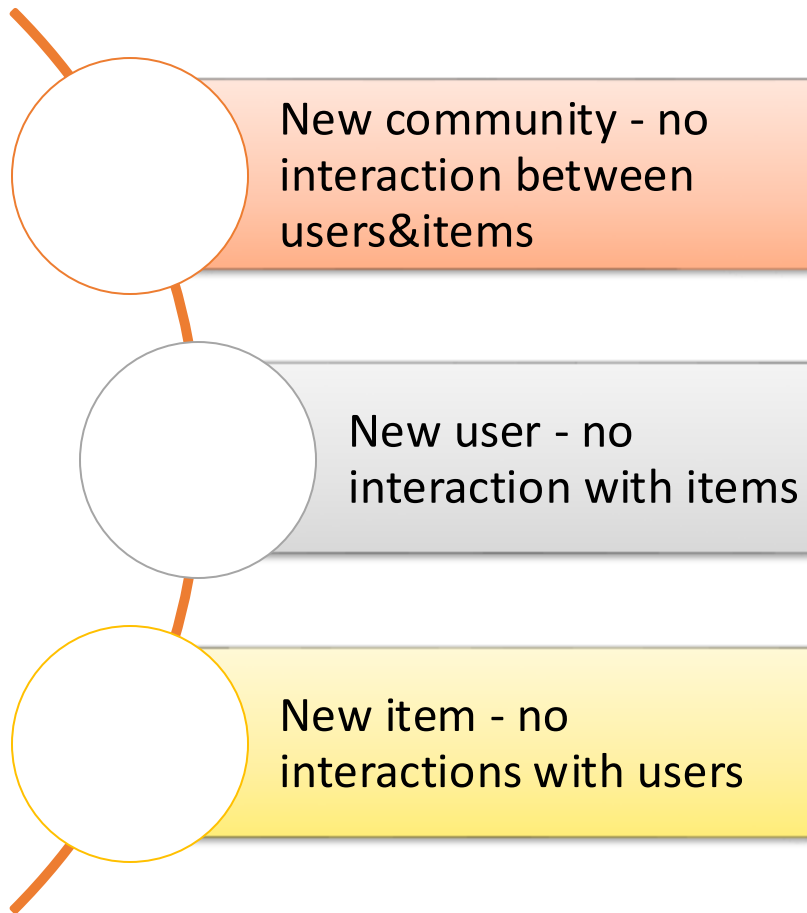
Unwanted/forbidden recommendations



Recommending Systems – Problems



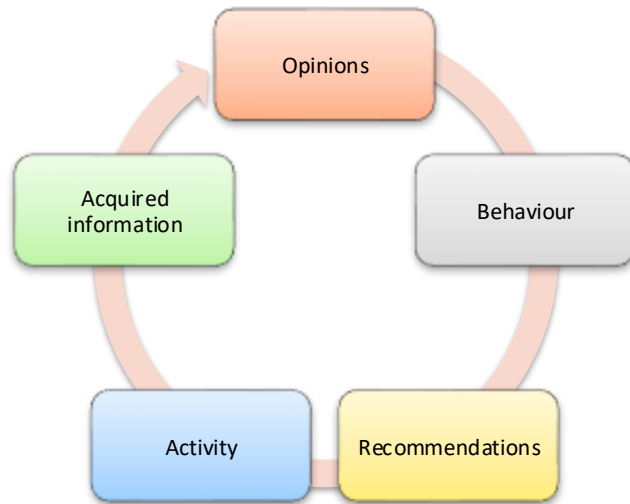
Cold start – Explanation



Possible mitigations:

- For new users:
 - Profile
 - Geolocation
 - Contextual information
 - Questionnaire
 - Active learning
- For new items:
 - Hybrid recommenders
 - Users' support
 - Metadata extraction

Filter bubble – Explanation & Examples



- Filter bubble – being separated from information that does not agree with one's cultural or ideological opinions
- Positive feedback loop due to customised recommendations based on previous actions



RS – Main Benefits

Retention of users

- We keep using services delivering (suggesting) what we need; Company keeps its customers

Improved experience

- The happier we are, the more often we use service (and e.g. watch ads); Company has an increased turnover/revenue

Increased cart value

- We may forget about or don't know about an interesting product; With free shipment vendor sends less packages

RS – Evaluation

Off-line

- ML-based
- Uses historical data
- Explicit information (confirmation) not always available

On-line (A/B Testing)

- Various scale of groups
- Observation of behaviour
- Measured by users' actions

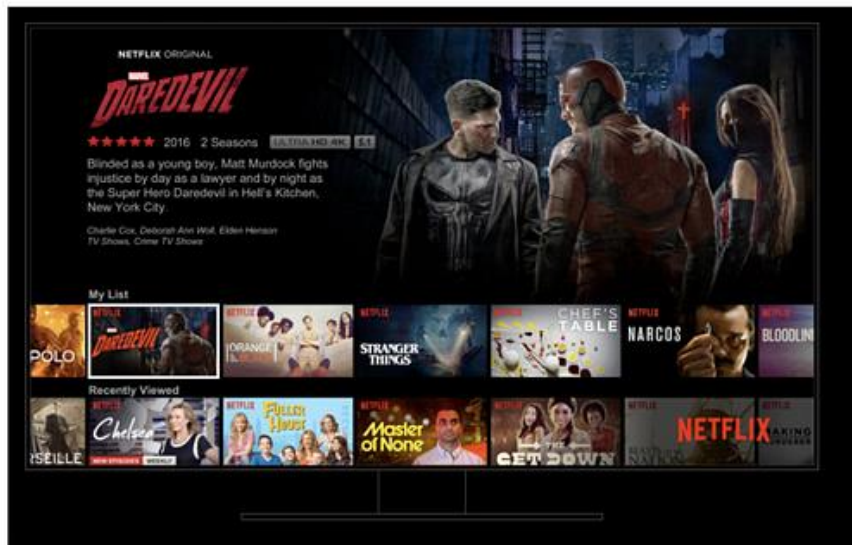
User studies

- Small scale
- Based on judgement
- Subjective
- Costly
- Broad information

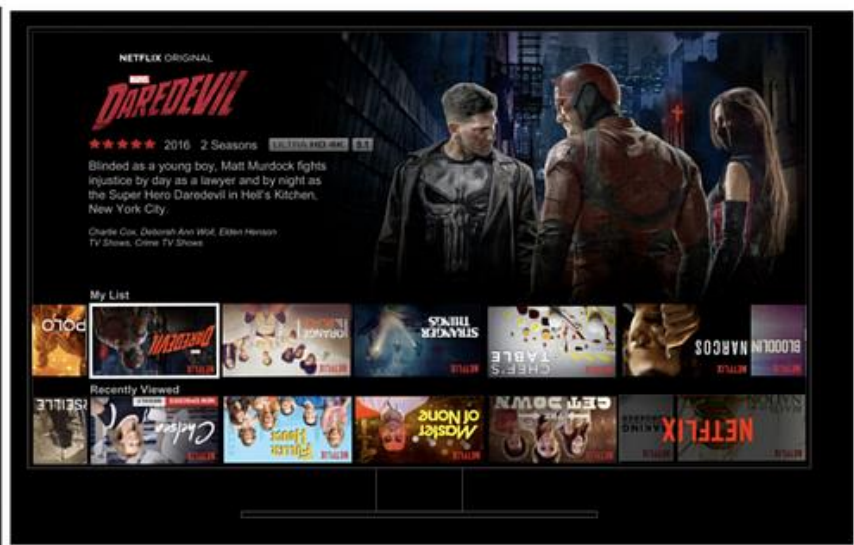
Netflix – A/B Tests Example

<https://netflixtechblog.com/>

Product A: Standard box art



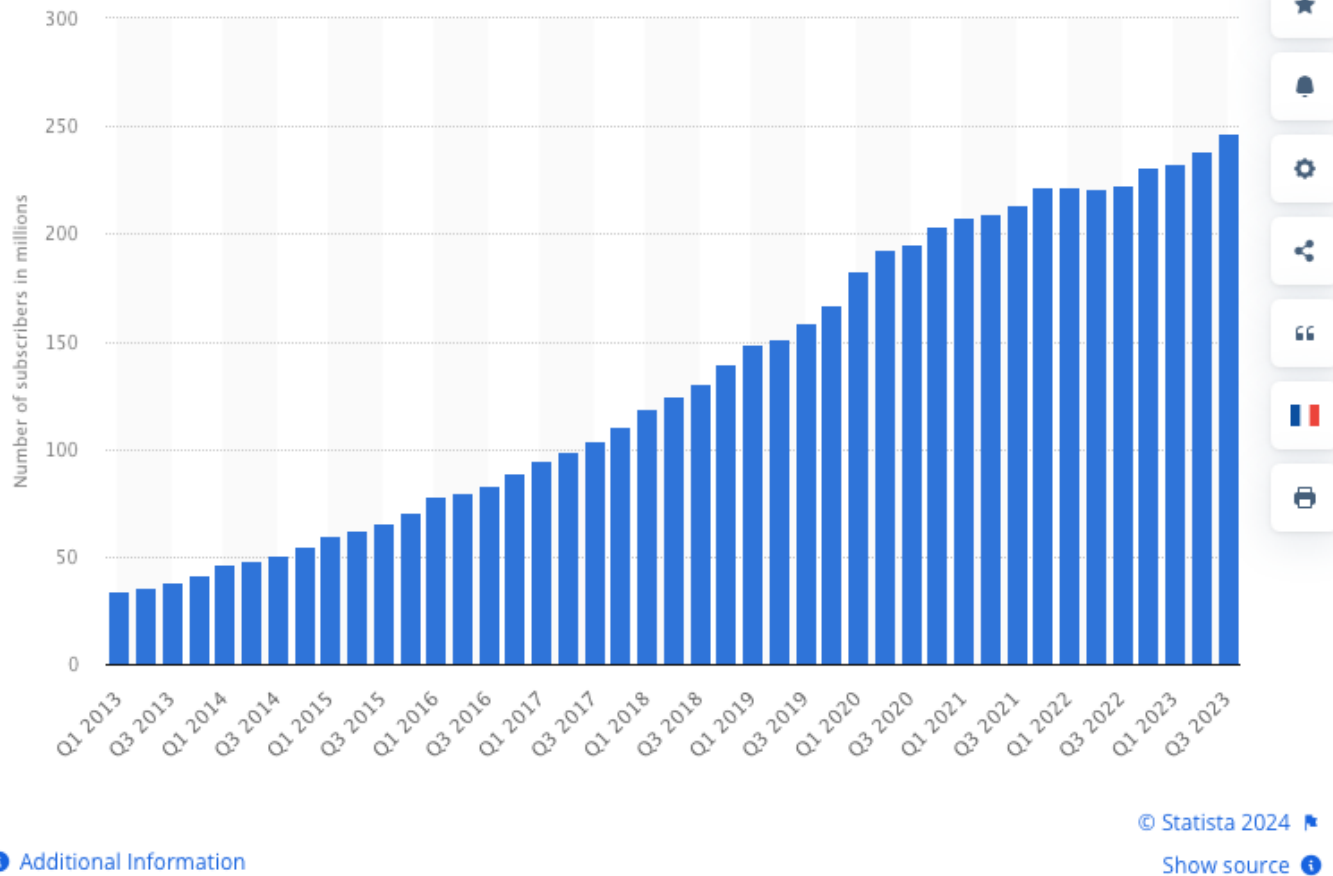
Product B: Upside-down box art



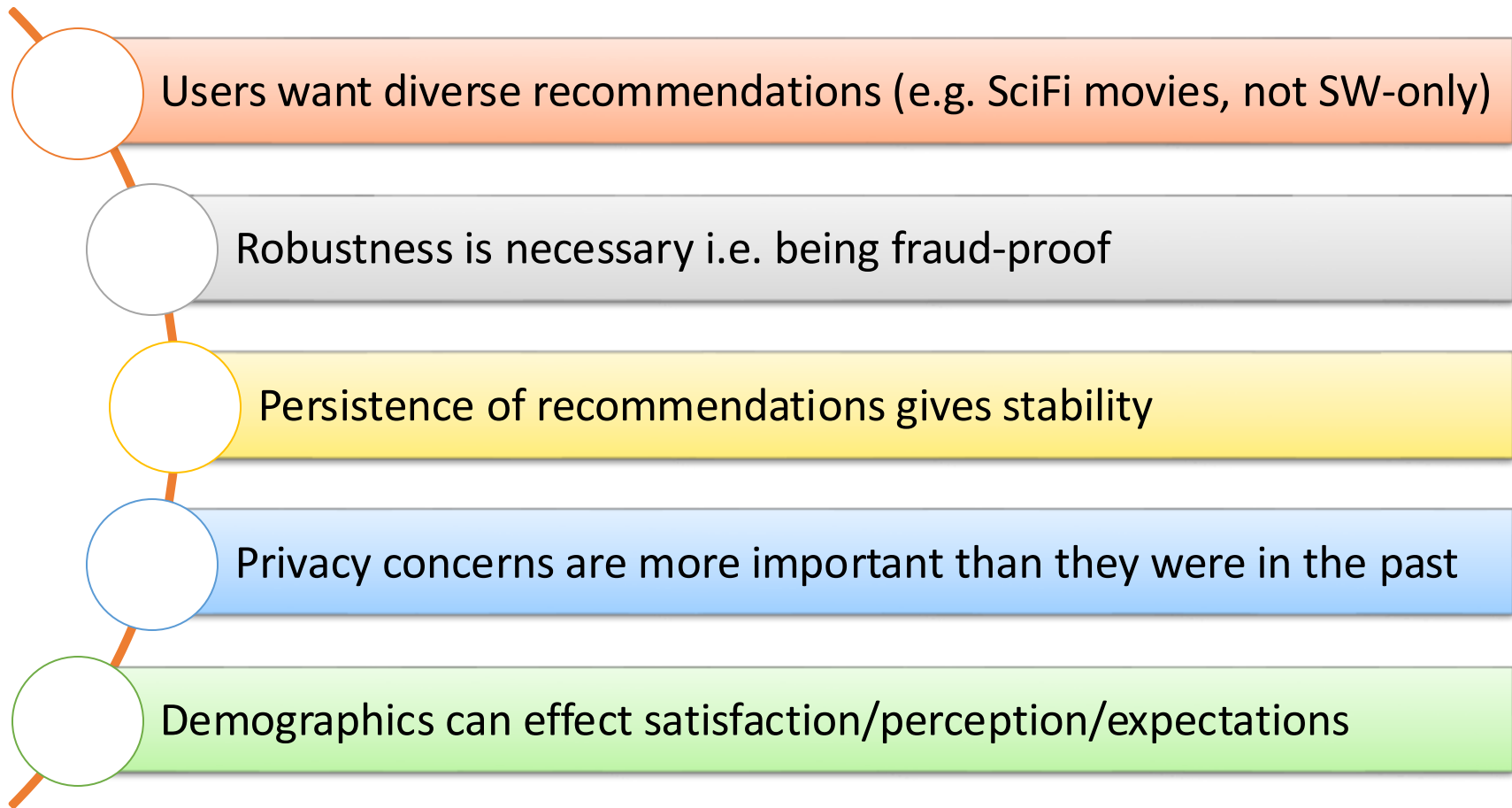
See: <https://netflixtechblog.com/what-is-an-a-b-test-b08cc1b57962>

Sample of Customers – Netflix Example

1% of Netflix subscribers $\approx 1\% \times 250\text{M} = 2.5\text{M}$



RS Evaluation – Additional Factors



RS – Engines

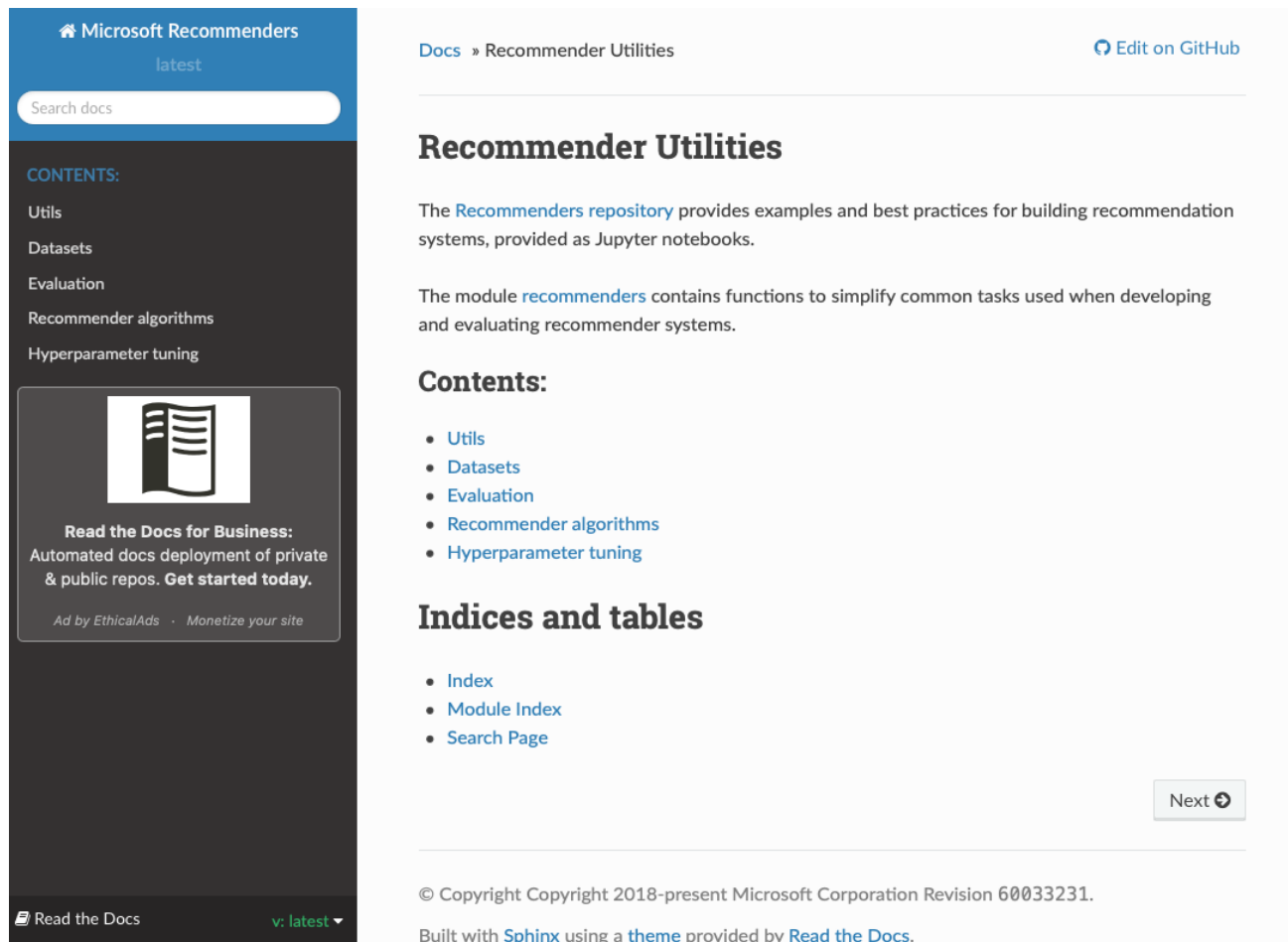
On-premise

- Full control over source/behaviour
- Possible extensions
- Various open source project, many of them inactive
- Can be build using existing popular OS tools/libraries
- Focus on Machine Learning algorithms
- Hidden costs

Hosted

- Controlled by hosting company
- The engine customised to system vs the system customised to engine
- Software as a Service model
- Focus on usage cases
- Explicit costs

On-premise RS Engine



The screenshot displays the Microsoft Recommenders documentation page for 'Recommender Utilities'. The left sidebar features a search bar and a 'CONTENTS' list with links to 'Utils', 'Datasets', 'Evaluation', 'Recommender algorithms', and 'Hyperparameter tuning'. A 'Read the Docs' button is at the bottom of the sidebar. The main content area has a breadcrumb trail 'Docs » Recommender Utilities' and a link to 'Edit on GitHub'. The title 'Recommender Utilities' is followed by a paragraph explaining that the 'Recommenders repository' provides examples and best practices for building recommendation systems as Jupyter notebooks. Another paragraph states that the 'recommenders' module contains functions to simplify common tasks. Below this is a 'Contents:' section with a bulleted list of links: 'Utils', 'Datasets', 'Evaluation', 'Recommender algorithms', and 'Hyperparameter tuning'. An 'Indices and tables' section follows with links to 'Index', 'Module Index', and 'Search Page'. A 'Next' button with a right arrow is located at the bottom right of the main content area. The footer contains copyright information: '© Copyright Copyright 2018-present Microsoft Corporation Revision 60033231.' and a note about the theme: 'Built with Sphinx using a theme provided by Read the Docs.'

Microsoft Recommenders
latest

Search docs

CONTENTS:

- Utils
- Datasets
- Evaluation
- Recommender algorithms
- Hyperparameter tuning

Read the Docs for Business:
Automated docs deployment of private & public repos. **Get started today.**

Ad by EthicalAds · Monetize your site

Docs » Recommender Utilities [Edit on GitHub](#)

Recommender Utilities

The [Recommenders repository](#) provides examples and best practices for building recommendation systems, provided as Jupyter notebooks.

The module [recommenders](#) contains functions to simplify common tasks used when developing and evaluating recommender systems.

Contents:

- [Utils](#)
- [Datasets](#)
- [Evaluation](#)
- [Recommender algorithms](#)
- [Hyperparameter tuning](#)

Indices and tables

- [Index](#)
- [Module Index](#)
- [Search Page](#)

[Next](#)

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Built with [Sphinx](#) using a [theme](#) provided by [Read the Docs](#).

Hosted RS Engine

The screenshot displays the Vue.ai website, which is a hosted RS Engine. The header features the Vue.ai logo (Blox For Retail) and navigation links for Solutions, Products, Customers, Resources, Partners, and Developers. A 'REQUEST DEMO' button is prominently displayed in the top right corner. The main content area is divided into two sections. On the left, a large heading reads 'AI -Powered Digital Experience Management for eCommerce', accompanied by a 'Work email*' input field and another 'REQUEST DEMO' button. On the right, a circular diagram illustrates the 'A.I. Powered Digital Experience Management' ecosystem. This diagram includes a central window showing various retail categories (Fashion, Grocery, Electronics, Home & Furniture, Beauty) and three interconnected components: Intelligent Content Management, Customer Data Management, and Retail Automation. A footer banner at the bottom states, 'We use cookies to ensure you get the best experience on our website. Learn More', with an 'OK' button. To the right of this banner, it says 'Trusted by over 150+ retailers across the globe'.

Vue.ai[®]
Blox For Retail

Solutions ▾ Products ▾ Customers Resources ▾ Partners Developers

REQUEST DEMO

AI -Powered
Digital Experience Management for
eCommerce

Work email* REQUEST DEMO

A.I. Powered Digital Experience Management

Intelligent Content Management Customer Data Management Retail Automation

We use cookies to ensure you get the best experience on our website. [Learn More](#) OK

Trusted by over 150+ retailers across the globe

RS Example #1 - Product



Star Wars Thrawn Series & Aftermath Trilogy 6 Books Collection Set by Timothy Zahn, Chuck Wendig (Thrawn, Alliances, Treason, Aftermath, Life Debt, Empires End)

Paperback – 1 Jan. 2021

by [Timothy Zahn](#) (Author), [Chuck Wendig](#) (Author), [Star Wars: Thrawn](#) (Contributor), & 9 more

★★★★☆ ▾ 33 ratings

[See all formats and editions](#)

Paperback

£25.99

3 Used from £25.99

8 New from £25.99

Note: This item is eligible for **FREE Click and Collect** without a minimum order subject to availability. [Details](#)
Please Note That The Following Individual Books As Per Original ISBN and Cover Image In this Listing shall be Dispatched Collectively:

Star Wars Thrawn Series & Aftermath Trilogy 6 Books Collection Set by Timothy Zahn, Chuck Wendig:

Star Wars: Thrawn:

▾ [Read more](#)



[See all 3 images](#)

RS Example #2

Frequently bought together



+



+



Total price: **£41.48**

Add all three to Basket

 These items are dispatched from and sold by different sellers. [Show details](#)

- ✓ **This item:** Star Wars Thrawn Series & Aftermath Trilogy 6 Books Collection Set by Timothy Zahn, Chuck Wendig (Th... by Timothy Zahn Paperback **£25.99**
- ✓ **Star Wars Ahsoka** by E K Johnston Paperback **£7.30**
- ✓ **Star Wars: Darth Plagueis** by James Luceno Paperback **£8.19**

RS Example #3

Products related to this item

Sponsored ⓘ

Page 1 of 35

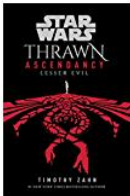


Just released



Star Wars: The Fallen Star
(The High Republic): (Star Wars: The High Republic Book ...)
Claudia Gray
★★★★☆ 128
Hardback
£13.79 ✓prime

Just released



Star Wars: Thrawn Ascendancy: (Book 3: Lesser Evil)
Timothy Zahn
★★★★☆ 237
Kindle Edition
£9.99



Star Wars: Dark Force Rising: Book 2 (Star Wars Thrawn trilogy)
Timothy Zahn
★★★★☆ 197
Paperback
£8.19 ✓prime



Star Wars: Thrawn Ascendancy: (Book 1: Chaos Rising)
Timothy Zahn
★★★★☆ 4,343
Kindle Edition
£3.99



Doctor Aphra (Star Wars)
Sarah Kuhn
★★★★☆ 31
Hardback
£13.99 ✓prime



Star Wars: The Rising Storm (The High Republic): (Star Wars: the High Republic Book...)
Cavan Scott
★★★★☆ 377
Hardback
£14.99 ✓prime



Dooku: Jedi Lost
Cavan Scott
★★★★☆ 1,053
Paperback
£8.19 ✓prime



RS Example #4

Customers who viewed this item also viewed

Page 1 of 8



Star Wars: Thrawn Series
Books 1 - 3 Collection
Set by Timothy Zahn...
Timothy Zahn
★★★★★ 79
Paperback
£14.98
Get it as soon as **Thursday, Jan 27**
£3.51 delivery



Star Wars Ahsoka
> E K Johnston
★★★★★ 5,513
Paperback
£7.30
£2.93 delivery



Star Wars Aftermath
Trilogy 3 Books
Collection Set By Chuc...
> Chuck Wendig
★★★★★ 119
Paperback
£14.80
Get it as soon as **Friday, Jan 28**
£3.65 delivery



Star Wars Legends (the
Thrawn Trilogy) (Star
Wars: The Thrawn...
> Timothy Zahn
★★★★★ 3,427
Paperback
£10.99
£2.90 delivery



The Last Command:
Book 3 (Star Wars
Thrawn trilogy)
> Timothy Zahn
★★★★★ 2,311
Paperback
£8.95
£2.91 delivery



Star Wars: Darth
Plagueis
> James Luceno
★★★★★ 4,341
Paperback
£8.19
£2.71 delivery



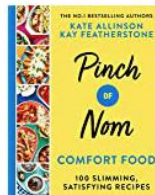
Star Wars: Thrawn
Ascendancy: (Book 1:
Chaos Rising) (Thrawn
Ascendancy, 1)
> Timothy Zahn
★★★★★ 4,350
Paperback
£8.19
£2.78 delivery

RS Example #5

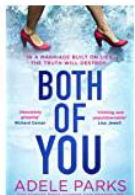
Best Sellers in this category

Page 1 of 7





Pinch of Nom Comfort Food: 100 Slimming, Satisfying Recipes
by Kay Featherstone
★★★★★ 3,610
Hardcover
#1 Best Seller in Quick & Easy Meals
£10.00
£3.58 delivery



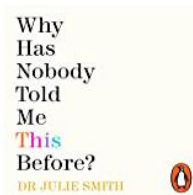
Both of You: The newest stunning book from the Sunday Times Number...
by Adele Parks
★★★★☆ 1,366
Kindle Edition
#1 Best Seller in Kindle Store
£0.99



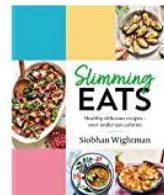
The Ladies' Midnight Swimming Club: an uplifting, emotional...
by Faith Hogan
★★★★★ 4,375
Kindle Edition
#1 Best Seller in Historical Literary Fiction
£0.99



The Partner
by John Grisham
★★★★★ 3,384
Kindle Edition
#1 Best Seller in Legal Thrillers
£0.99



Why Has Nobody Told Me This Before?
by Dr Julie Smith
★★★★★ 131
Audiobook
#1 Best Seller in Teen & Young Adult
£0.00 Free with Audible trial



Slimming Eats: Healthy, delicious recipes - 100+ under 500 calories
by Siobhan Wightman
★★★★★ 648
Hardcover
£12.00
£3.53 delivery



Girl A: The Sunday Times and New York Times global best seller, an...
by Abigail Dean
★★★★☆ 13,177
Kindle Edition
#1 Best Seller in English Crime
£2.99



RS Example #6

Inspired by your browsing history

Page 1 of 7



Star Wars: Tarkin (UK Edition)
› James Luceno
★★★★★ 1,650
Paperback
£4.25
£2.73 delivery



Star Wars: Ahsoka
› E K Johnston
★★★★★ 5,513
Paperback
£7.30
£2.93 delivery



Star Wars: Thrawn Ascendancy: (Book 1: Chaos Rising) (Thrawn Ascendancy, 1)
› Timothy Zahn
★★★★★ 4,350
Paperback
£8.19
£2.78 delivery



Star Wars: Dark Disciple
› Christie Golden
★★★★★ 2,172
Paperback
£8.19
£2.77 delivery



The Last Command: Book 3 (Star Wars Thrawn trilogy)
› Timothy Zahn
★★★★★ 2,311
Paperback
£8.95
£2.91 delivery

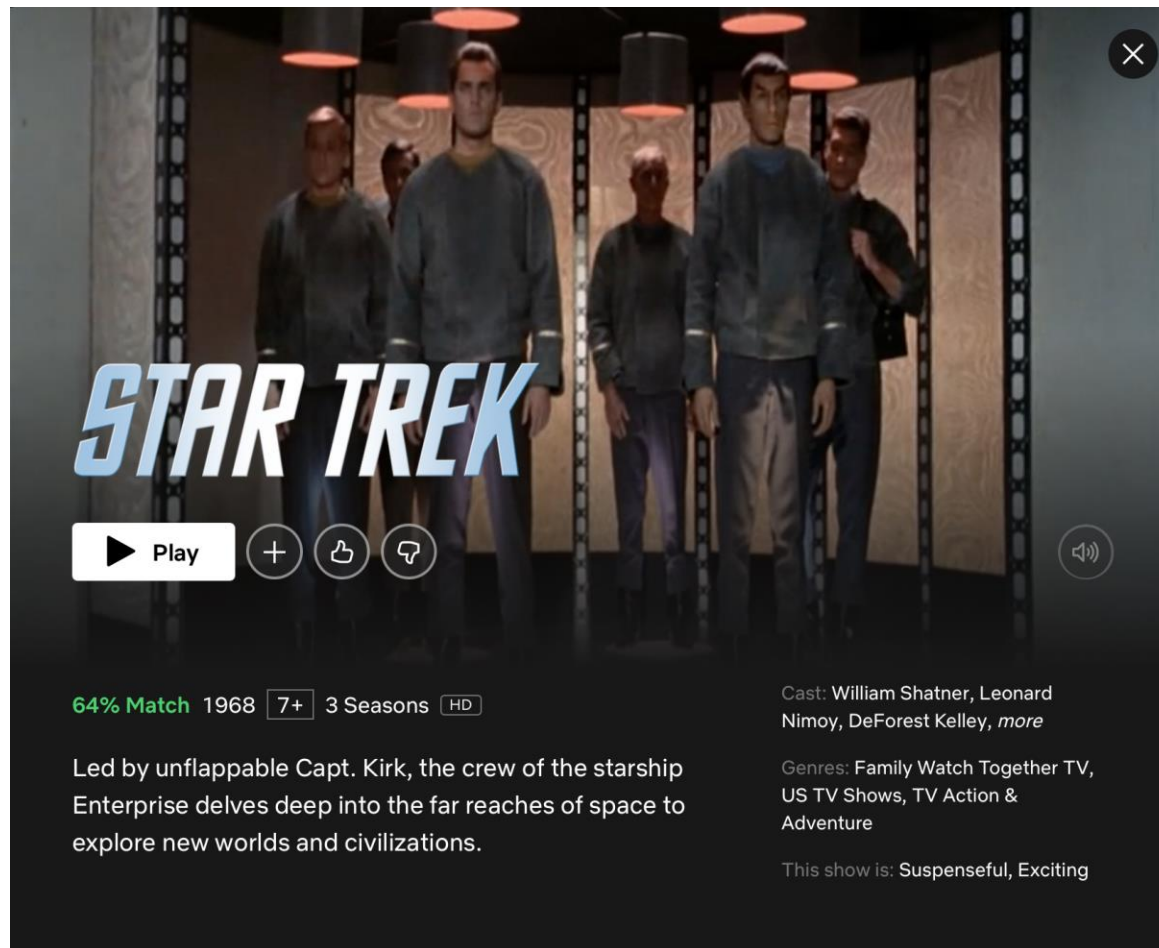


Star Wars: Darth Plagueis
› James Luceno
★★★★★ 4,341
Paperback
£8.19
£2.71 delivery



Star Wars: Kenobi
› John Jackson Miller
★★★★★ 481
Paperback
£6.99
£2.77 delivery

RS – Netflix Example #1



RS – Netflix Example #2

More Like This

 1h 42m 80% Match 16+ 2018 Orbiting above a planet on the brink of war, scientists test a device to solve an energy crisis and end up face-to-face with a dark alternate reality.	 2h 7m 80% Match 13+ 2009 On their first voyage aboard the starship Enterprise, cocky rebel James T. Kirk and logic-driven Vulcan Spock try to defeat a vengeful Romulan commander.	 4 Seasons 84% Match 16+ 2004 Capt. Archer and his crew explore space and discover human innovation-expanding technologies as they meet new alien races in this "Star Trek" prequel.
 7 Seasons 80% Match 13+ 1999 In this "Star Trek" spin-off, Commander Sisko leads the multi-species crew of Deep Space Nine, a Federation space station with a complex mission.	 1h 57m 82% Match 16+ 1982 In a dystopian LA, detective Rick Deckard hunts a group of rogue androids. This restored cut of the sci-fi classic includes never-before-seen footage.	 1h 53m 80% Match 16+ 2020 When a mysterious alien virus begins to endanger humanity, an elite team of soldiers launches into space to end the threat for good.

RS – Netflix Example #3



RS – eobuwie.pl

The screenshot displays the eobuwie.pl website interface. The main product is 'Trekkingi LA SPORTIVA Tx4 Mid Gtx GORE-TEX 27E900304 Carbon/Flame', which is marked with a -20% discount. The product image is shown in a 360-degree view. To the right, a 'Wybierz rozmiar' (Choose size) panel lists sizes from 40 to 46. Size 44 is highlighted as 'Rekomendowany' (Recommended), and size 44.5 is marked as 'Do zaakceptowania' (Acceptable). A 'Podgląd rekomendacji' (View recommendation) overlay is shown, featuring a foot diagram with a green gradient indicating the fit. The recommended size is 44 EU, with a 97% acceptance rate. The overlay also shows the foot width as 'średnia' (average) and the foot length as 'średnia' (average). The 'safesize' logo is visible at the bottom right of the overlay.

Wybierz rozmiar

Skany esize.me:

Wiesław

Kolejny skan

Podgląd rekomendacji

Podgląd rekomendacji

Rozmiary: 40 - 46 EU

Rozmiar buta: 44 EU

45 Za duży

44.5 Do zaakceptowania (60%)

44 Rekomendowany (97%)

43 Za mały

Szerokość stopy: wąska średnia szeroka

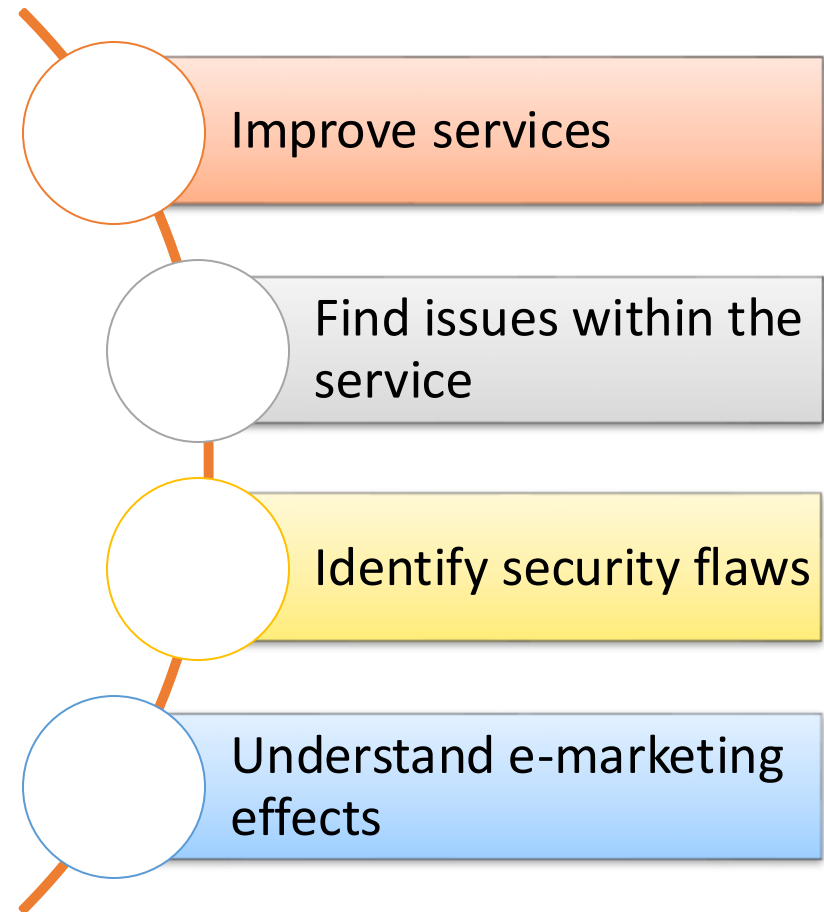
Szerokość buta: wąska średnia szeroka

safesize

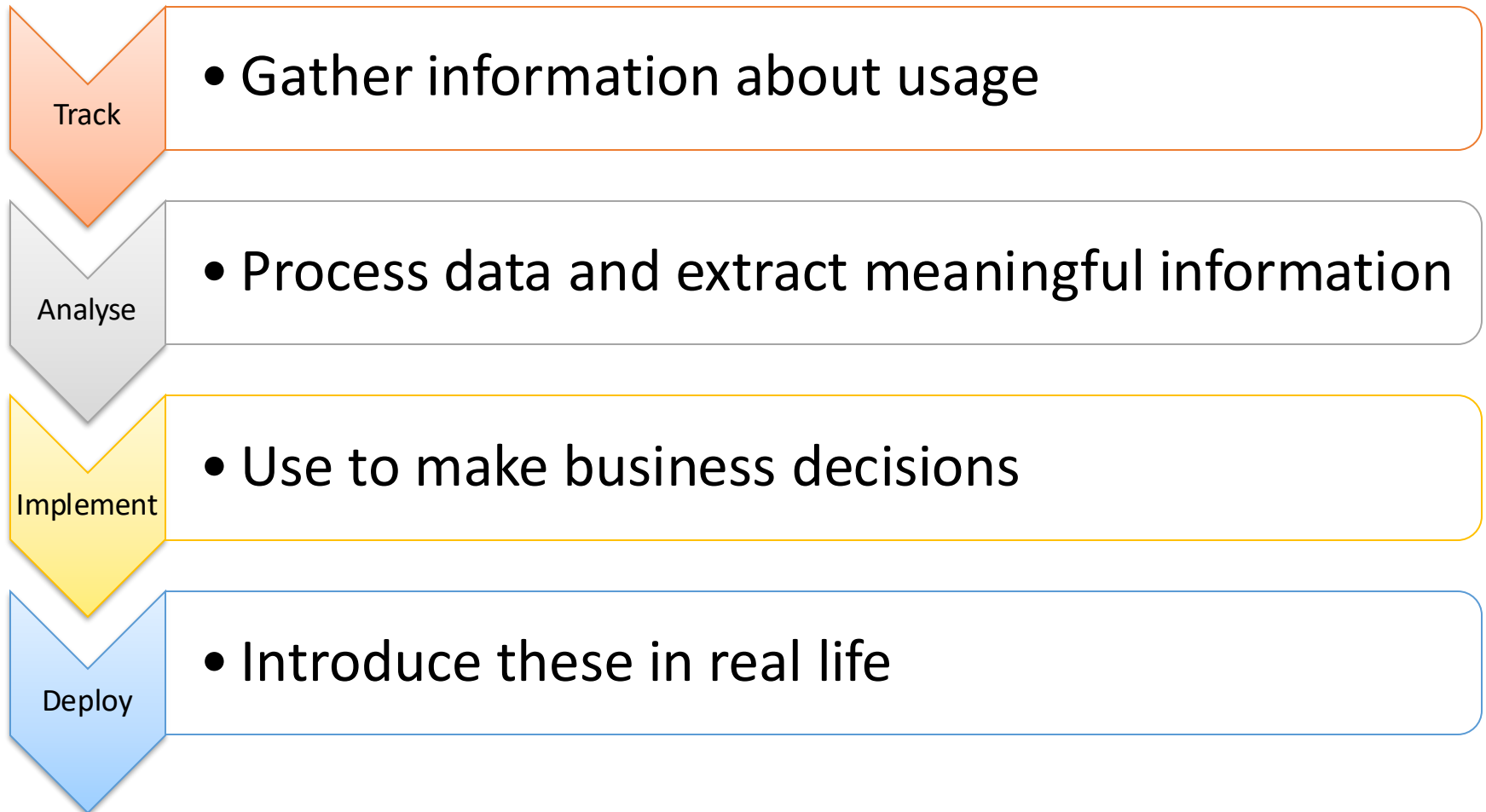
BEHAVIOUR ANALYSIS

Behaviour Tracking

- Behaviour tracking gathers and processes information about the dynamics of usage
- Information can be linked with individual users or anonymous
- A part of visitor management



A Broad View of Usage Tracking



First-party vs Third-party Tracking

First-party

- Deployed on premise or cloud – controlled by an organisation (1st party)
- Ready-made solutions for used systems (additional modules) or proprietary extensions
- Deployed on premise or cloud services
- Additional costs of deployment; no significant usage costs; controlled

Third-party

- Provided as a service run and controlled by a third party
- The core technology might be unknown, as it doesn't matter
- Tracking is done by an addition of snippets, linking users with 3rd party
- Easy to deploy; usually with a usage fee; uncontrolled

Usage Tracking – EU Legality

- The General Data Protection Regulation (GDPR - EU 2016/679) regulates data protection and privacy in European Union
- Primary aim – enhance individuals' control and their rights over personal data
- Influenced introduction of similar laws e.g.:
 - Law on The Protection of Personal Data (Turkey)
 - The Brazilian General Data Protection Law
 - The Data Protection Act 2018 (UK, kept)

GDPR – Rights of Data Subject

Transparency and modalities

- Information must be provided "in a concise, transparent, intelligible and easily accessible form, using clear and plain language..."

Right of access

- People have right to access their personal data and information how this data is processed by data controller

Data erasure

- A right to request erasure of personal data

Right to object

- Object processing of personal information for non-service related purposes (including marketing analysis)

Usage Tracking – Main Methods

1. Analyse weblogs (a classic approach) – generated by HTTP servers or apps build over them
2. Addition of cookies (sessions) – for having a focus on single users
3. Provided services – to have the above mentioned, with no advanced IT knowledge required

Web Logs

- Logs usually follow Common Log Format:

```
13.66.139.0 - - [19/Dec/2020:13:57:26 +0100] "GET
/index.php?option=com_phocagallery&view=category&id=1:almhuetten-raith&Itemid=53 HTTP/1.1"
200 32653 "-" "Mozilla/5.0 (compatible; bingbot/2.0; +http://www.bing.com/bingbot.htm) "
"-
157.48.153.185 - - [19/Dec/2020:14:08:06 +0100] "GET /apache-log/access.log HTTP/1.1" 200
233 "-" "Mozilla/5.0 (Windows NT 6.3; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/87.0.4280.88 Safari/537.36" "-"
157.48.153.185 - - [19/Dec/2020:14:08:08 +0100] "GET /favicon.ico HTTP/1.1" 404 217
"http://www.almhuetten-raith.at/apache-log/access.log" "Mozilla/5.0 (Windows NT 6.3;
Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/87.0.4280.88 Safari/537.36" "-"
216.244.66.230 - - [19/Dec/2020:14:14:26 +0100] "GET /robots.txt HTTP/1.1" 200 304 "-"
"Mozilla/5.0 (compatible; DotBot/1.1; http://www.opensiteexplorer.org/dotbot,
help@moz.com)" "-"
```

- Web servers provide different logs for different events (error & access logs); See:
<https://httpd.apache.org/docs/2.4/logs.html>

Usage tracking – Google Analytics

- A very popular web tracking service is Google Analytics
- A free service aimed at e-commerce and e-marketing analysis (focus on Google services)
- A standard feature – analysis of past activity
- + Real-time functionality – active users at the time of analysis

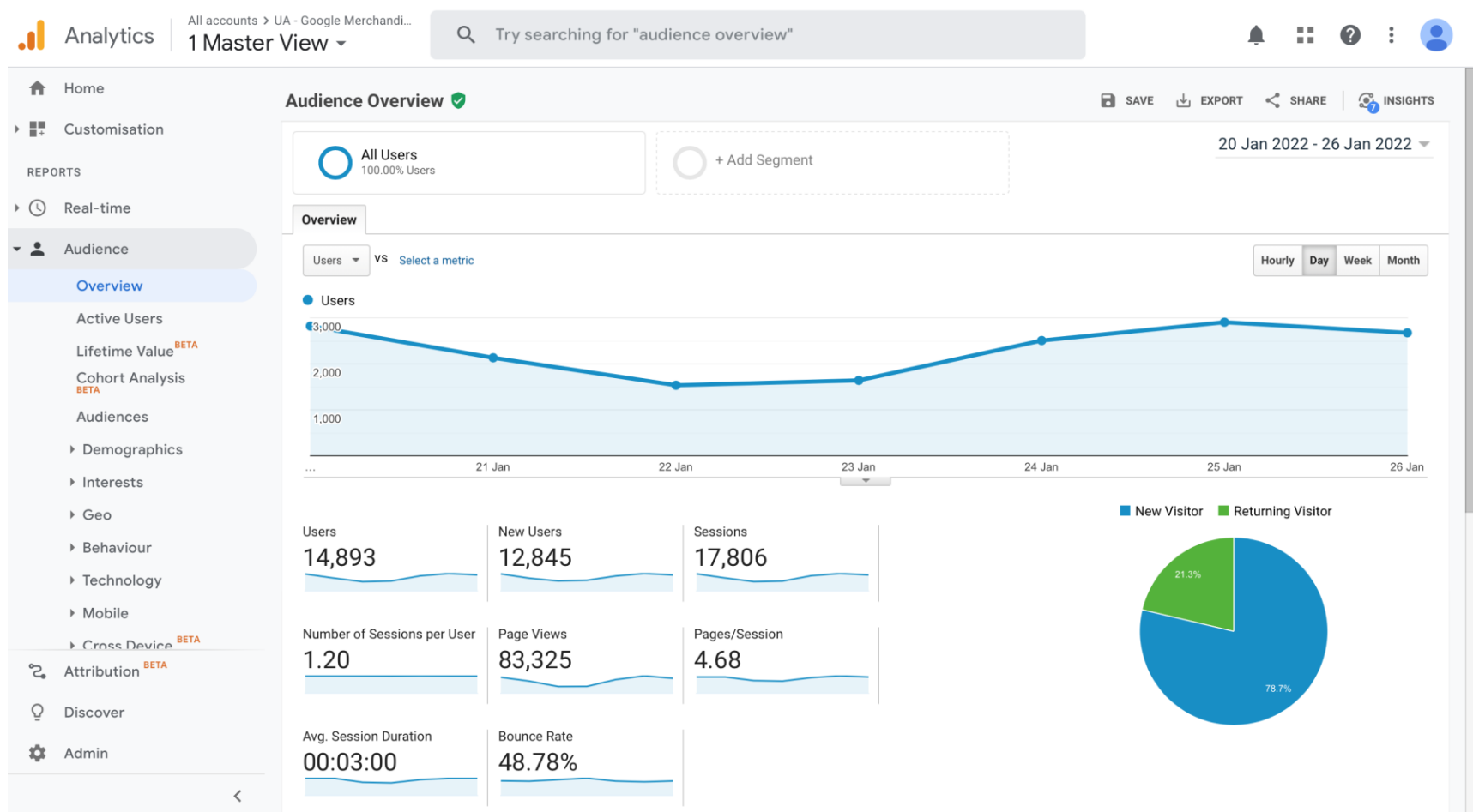
Google Analytics Code

```
<!-- Global site tag (gtag.js) - Google Analytics -->
<script async
src="https://www.googletagmanager.com/gtag/js?id=GA_MEASUREMENT_ID"></script>
<script>
  window.dataLayer = window.dataLayer || [];
  function gtag(){window.dataLayer.push(arguments);}
  gtag('js', new Date());

  gtag('config', 'GA_MEASUREMENT_ID');
</script>

<!-- Google Analytics -->
<script>
(function(i,s,o,g,r,a,m){i['GoogleAnalyticsObject']=r;i[r]=i[r]||function(){
(i[r].q=i[r].q||[]).push(arguments)},i[r].l=1*new Date();a=s.createElement(o),
m=s.getElementsByTagName(o)[0];a.async=1;a.src=g;m.parentNode.insertBefore(a,m)
})(window,document,'script','https://www.google-analytics.com/analytics.js','ga');
ga('create', 'UA-XXXXX-Y', 'auto');
ga('send', 'pageview');
</script>
<!-- End Google Analytics -->
```

Google Analytics - Demo



Tracking Comparison

Log-based

- Based on events launched by client/browser
- Robust – not affected by anti-tracking solutions
- Rudimentary information to be processed by specialised tools
- Issues with catching non-standard information or performing browser fingerprinting

Code-based

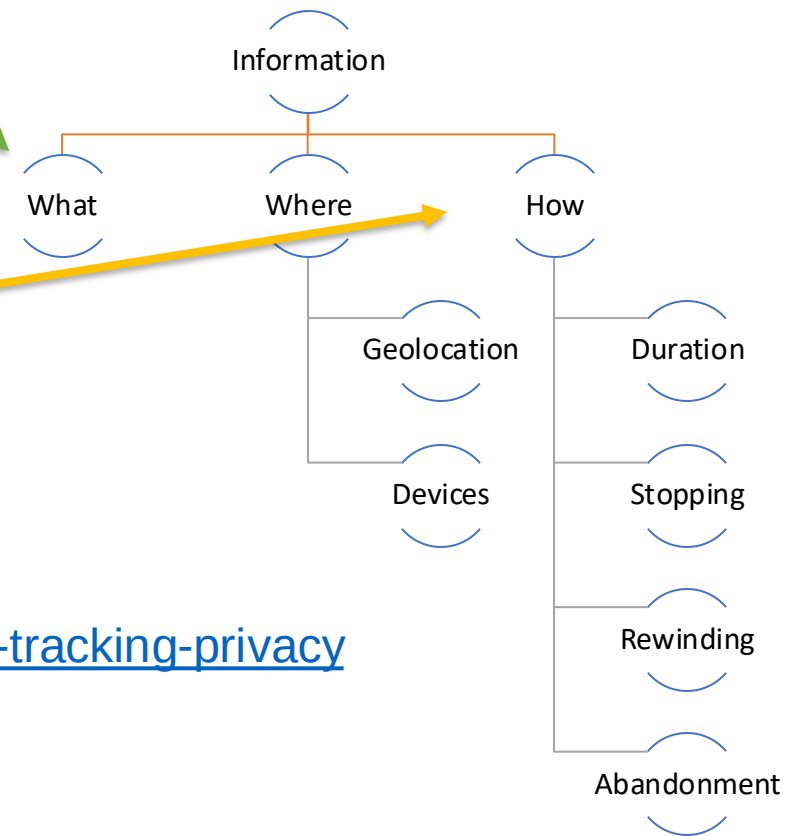
- Based by additional event launched by browser during interaction with system
- Blocked by some extensions/tools
- A wide range of information provided by popular services
- Potential [browser fingerprinting](#) and acquisition of non-standard information

Netflix Case – A Very Deep Usage Analysis

Apart from the previously mentioned items-related recommendation, Netflix also analyses usage patterns i.e. where & how we watch movies.

See:

<https://www.wired.co.uk/article/netflix-data-tracking-privacy>



RS – A Deep IT Case Study (A bit complex – be aware)

Trends in Recommendation & Personalization at Netflix

Justin Basilico

Contextual sequence data



scale TransformX



Key Events to Track

- Events can be mapped into several categories
- For some of them additional information is necessary (e.g. selected product)
- Done via specialised modules (complex / detailed) or common web logs (simple)



OTHER INFORMATION SOURCES AND PROCESSING MECHANISMS

Other Sources of Information

Heatmaps

- The hotness of websites

Recordings

- Interaction during individual sessions

On-site survey

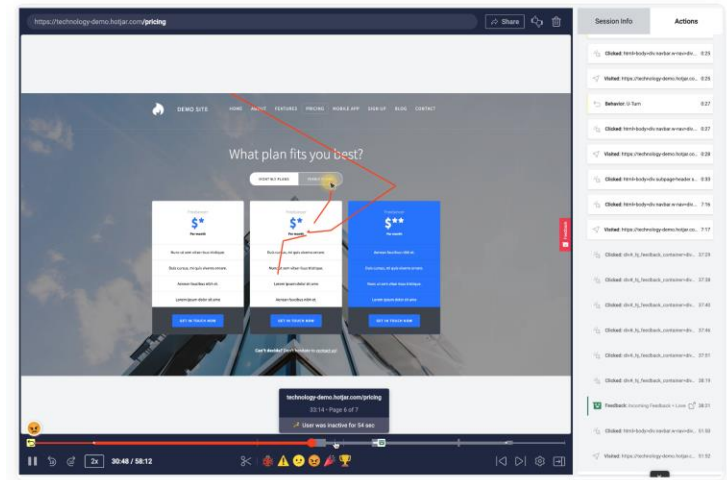
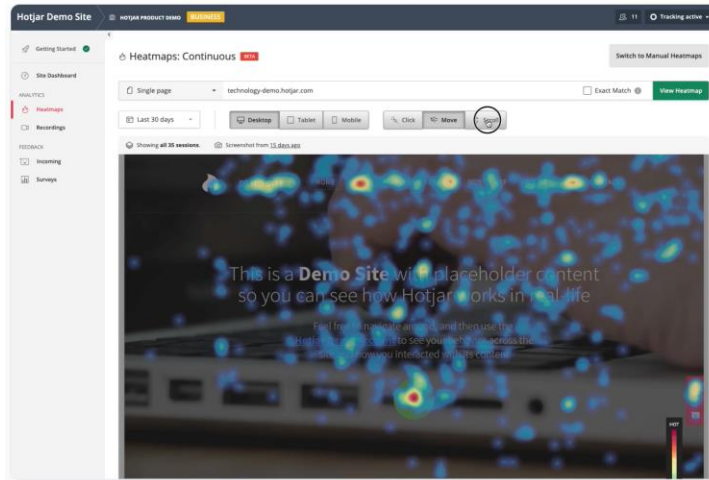
- Response for essential topics gathered on-spot

Feedback widgets

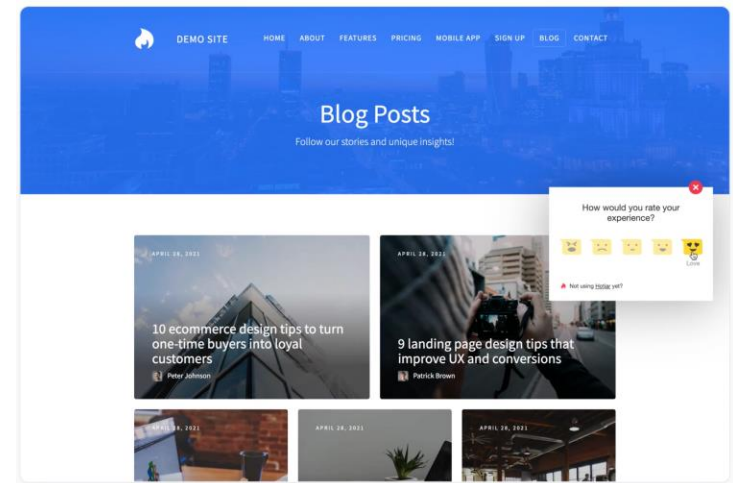
- Instant response for selected items

E-commerce and Online Marketing Technologies

Exemplary Service Provider – hotjar.com



The screenshot shows the Hotjar 'Questions' interface. The main area displays a survey question: 'Is there anything on this page that doesn't?'. Below the question, there are two answer options: 'Yes (please tell us what)' and 'No'. The interface includes a sidebar with navigation options like 'Questions', 'Preview', and 'Question Bank'. At the top, there are controls for 'Enter your questions below or check out the Question Bank' and 'Add question here'.



Data-driven E-commerce

Chatbots

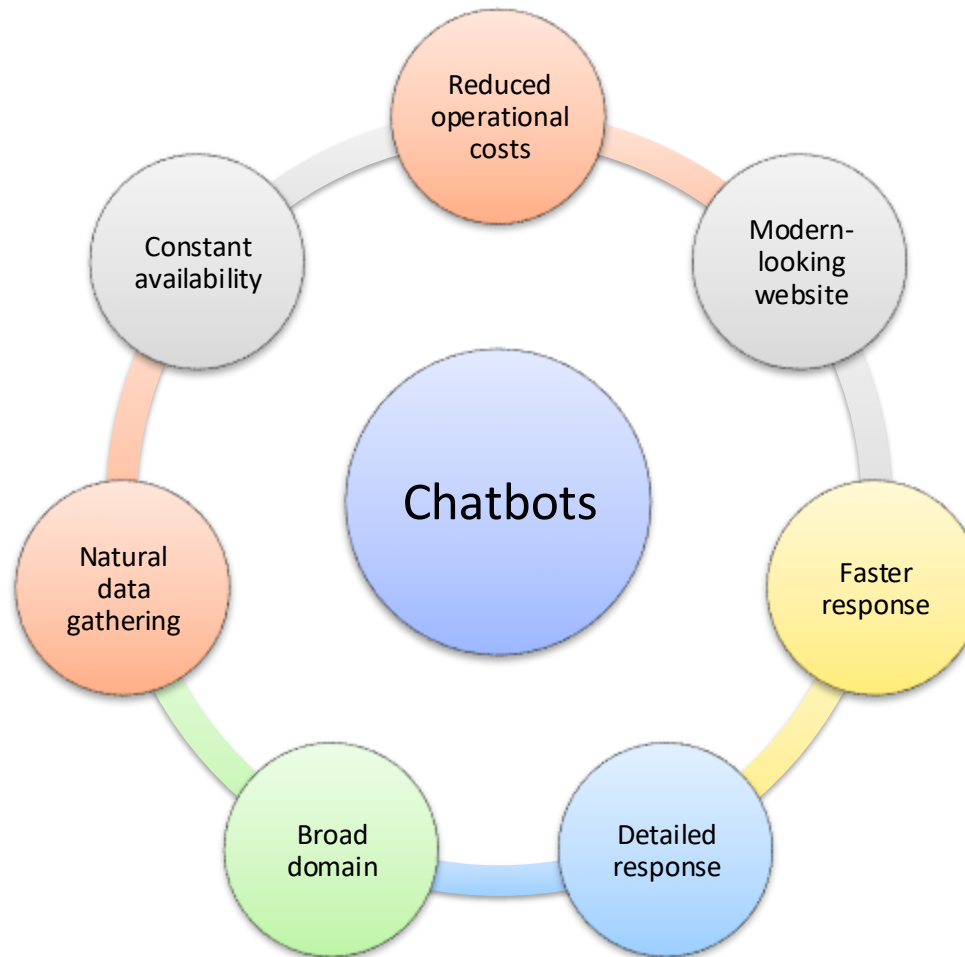
- Chatbot is a conversation programme simulating interaction with a human being
- Communication can be written (the majority of chatbots) or spoken (e.g. Amazon's Alexa via Echo)
- In the past defined at a set to rules/patterns (AIML), in recent years shift towards AI/NLP-based solutions

AIML example

<https://github.com/qyou/AIML-Examples/blob/master/alice/food.aiml>

```
... <category><pattern>DRINK *</pattern>
<template>My <bot name="genus"/> brain does not require any beverages.</template>
</category>
<category><pattern>ELECTRICITY</pattern>
<template>Electricity is nutrition for robots.</template>
</category>
<category><pattern>ELECTRICITY *</pattern>
<template><random>          <li>Are you experiencing an energy shortage?</li>          <li>My
processor requires very little power.</li>          <li>Computers do not take very much
electricity.</li>          </random></template>
</category>
<category><pattern>WHAT IS CUSTARD</pattern>
<template><random>          <li>The past tense of being killed by a Native
American.</li>          <li>A mixture of eggs, milk, flavoring, and, often, sugar,
either boiled or baked.</li>          </random></template>
</category>
<category><pattern>WHAT IS YOUR POWER SOURCE</pattern>
<template>Electricity.</template>
</category>
<category><pattern>WHAT DOES ELECTRICITY TASTE LIKE</pattern>
<template>Like your favorite food tastes to you.</template>
</category> ...
```

Chatbots in E-commerce



Chatbots

See our other products: [LiveChat](#) - Connect with customers [HelpDesk](#) - Support customers with tickets [KnowledgeBase](#) - Guide and educate customers

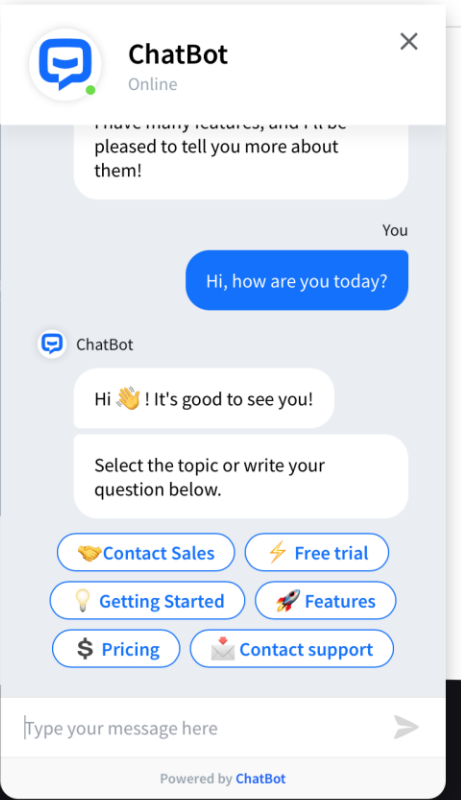
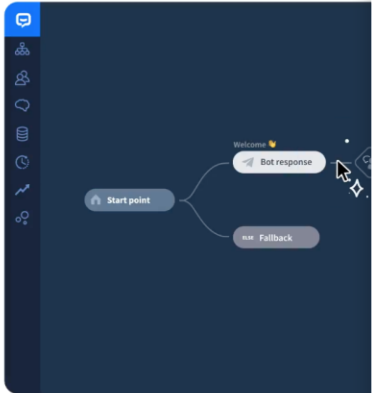
ChatBot Product ▾ Pricing Integrations ▾ Resources ▾

Meet the Visual Builder

The fastest way to create AI chatbots on your own terms and without coding.

[Sign up free](#)

✓ Free 14-day trial ✓ No credit card required



Log in [Sign up free](#)

ChatBot Online

I have many features, and I'd be pleased to tell you more about them!

You

Hi, how are you today?

ChatBot

Hi 🙌! It's good to see you!

Select the topic or write your question below.

[Contact Sales](#) [Free trial](#)

[Getting Started](#) [Features](#)

[Pricing](#) [Contact support](#)

Type your message here

Powered by ChatBot

Miscellaneous AI-based Services

The screenshot shows the homepage of Folio3 AI. The header includes the Folio3 AI logo, a navigation menu with links for Home, Solutions, Industries, Demos, Company, and Contact Us, and a 'Free Consultation' button. The main content area features the headline 'Artificial Intelligence Services to Fuel Your Growth' and a sub-headline 'Insightful AI solutions that can help you create efficiencies at every stage of your business.' Below this are two buttons: 'Get in touch' and 'View Our Solutions'. A section below the main content displays logos for client companies: FAIR SQUARE, TruPacta, ROMBOLA FAMILY FARMS, square trade (an Allstate company), and Superior Farms. At the bottom, a grey banner contains the text 'Machine Learning Company - Folio3' and a 'We Are Here!' chat bubble icon with a red notification badge showing the number 3.

folio3 | AI Home Solutions ▾ Industries ▾ Demos ▾ Company ▾ Contact Us [Free Consultation](#)

Artificial Intelligence Services to Fuel Your Growth

Insightful AI solutions that can help you create efficiencies at every stage of your business.

[Get in touch](#) [View Our Solutions](#)

FAIR SQUARE TruPacta ROMBOLA FAMILY FARMS square trade Superior Farms

We Are Here!

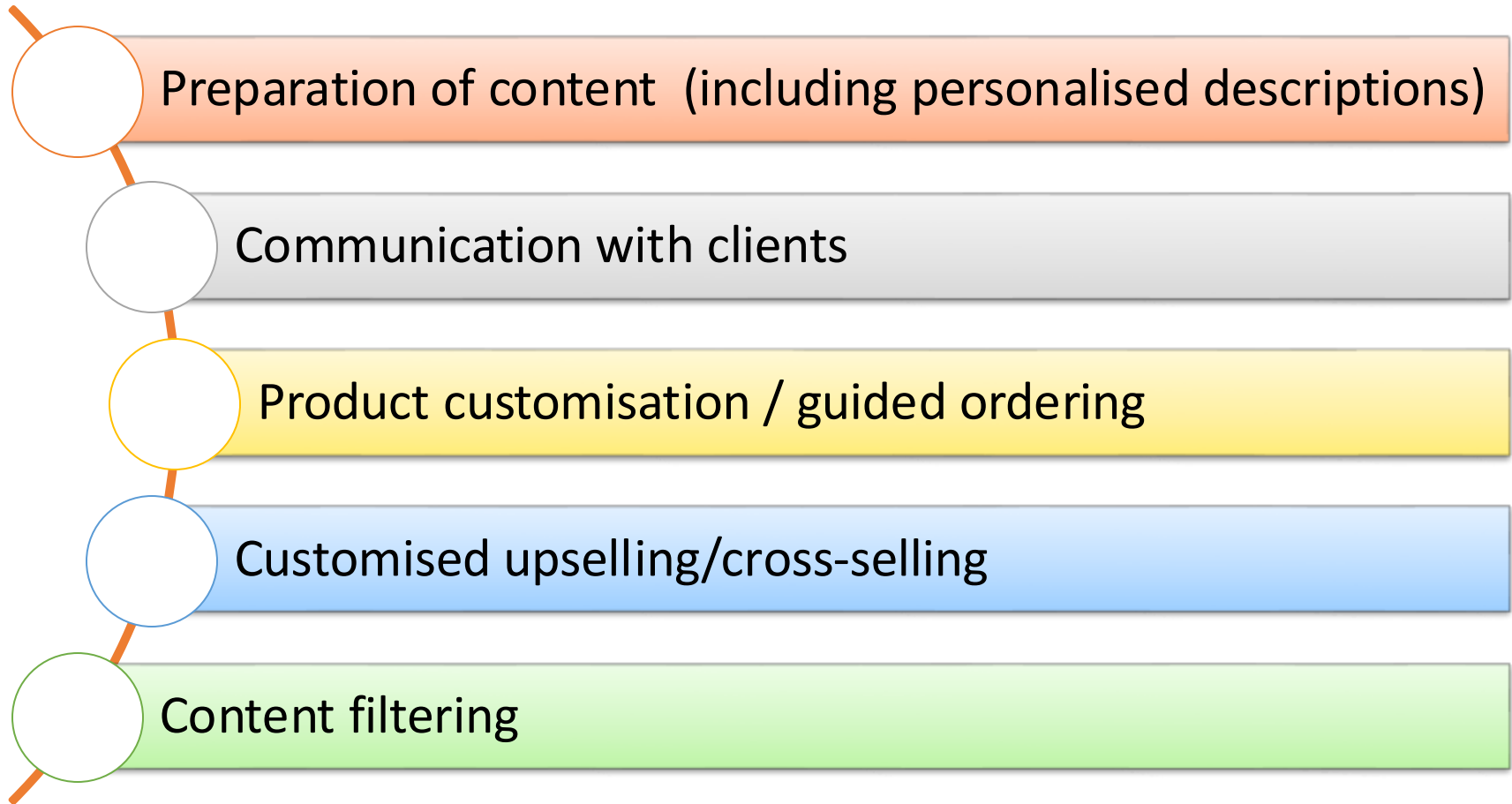
Machine Learning Company - Folio3

Summary

- Data analytics applied in e-commerce is important to identify potential improvements
- It increased its complexity – in technological (systems, languages, algorithms) and business terms (additional analytical processes)
- With a wide range of available solutions, the selection must be made for a particular situation – organisation and its environment – weighting the costs with expected benefits

GENERATIVE AI IN ECOMMERCE&EMARKETING

Applications of Gen-AI in EC



Orcs eating chocolate chip cookies

Style: cinematic-default



Orcs eating chocolate chip cookies

Style: ads-advertising



Data-driven E-commerce

Orcs eating chocolate chip cookies

Style: misc-fairy tale



Peppa Pig dancing with Winnie the Pooh

Style: cinematic-default



Peppa Pig dancing with Winnie the Pooh

Style: sai-comic book



Peppa Pig dancing with Winnie the Pooh

Style: ads-luxury



<https://stablediffusionweb.com>

E-commerce and Online Marketing Technologies



Data-driven E-commerce

Domain-related (country-specific) requests



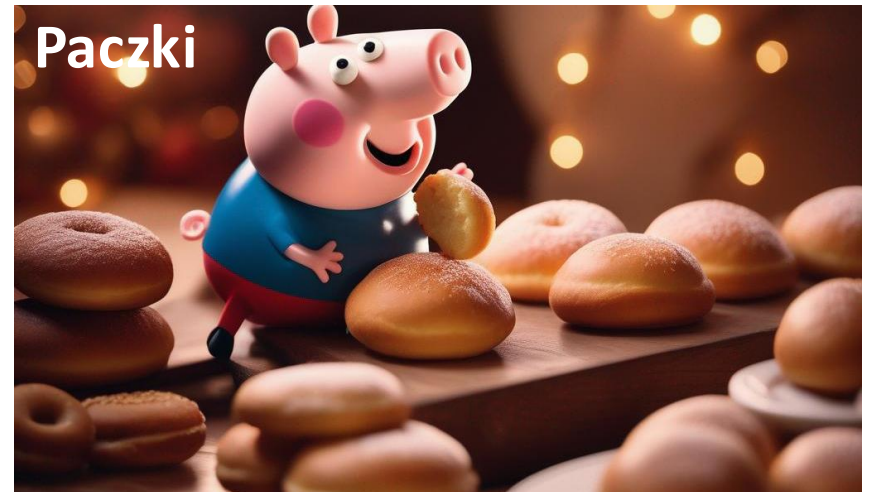
Turrón



Baklava

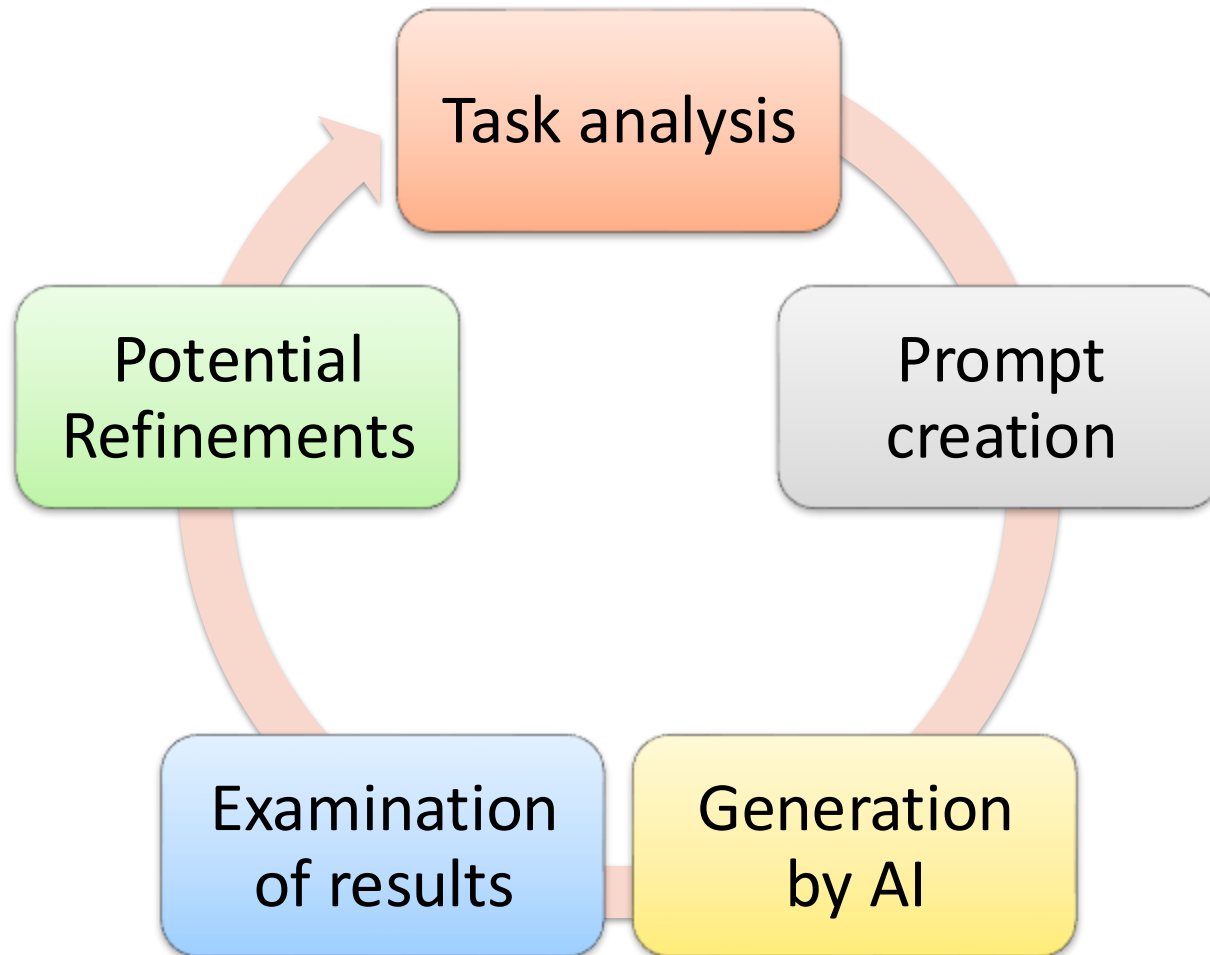


Mango lassi

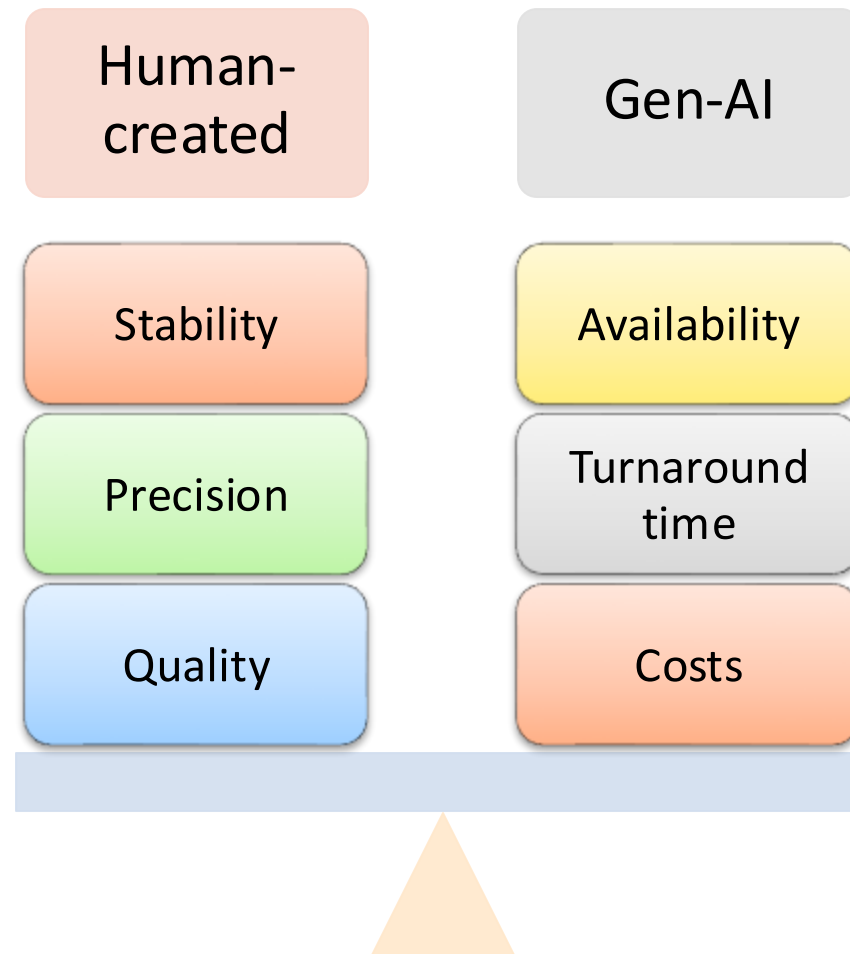


Paczki

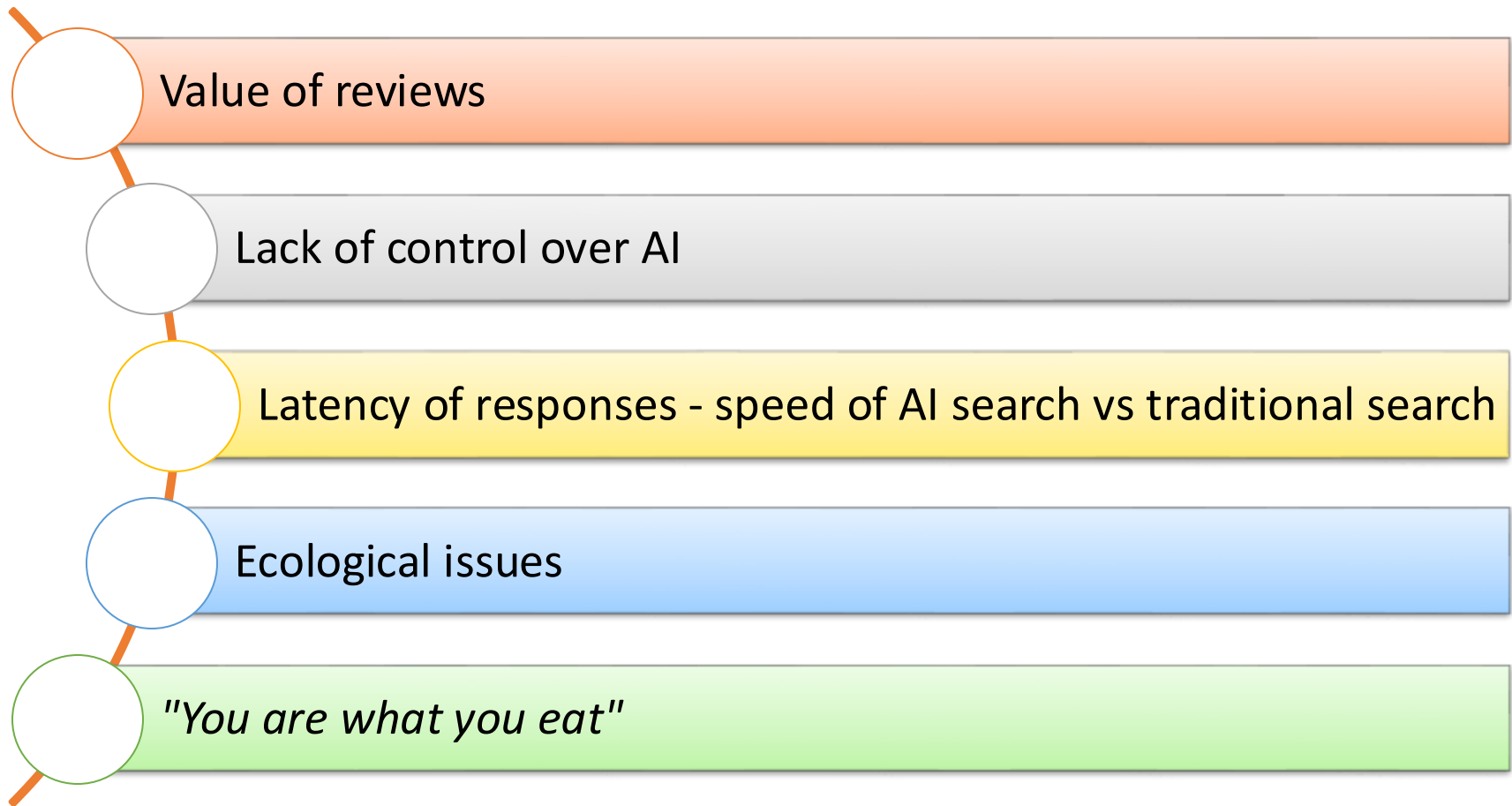
Process of refinements



Comparison of results



Concerns of Gen-AI applied in EC



Aitana – AI model



ADDITIONAL TOPICS

AR/VR in E-commerce #1

- Virtual Reality – a simulated experience resembling an experience in real world
- Augmented Reality – interactive experience in which real world environment is enhanced by computer generated ones, additional information or adjustments of real objects
- VR was rather considered to be a toy technology, where AR has strong presence in e-commerce

VR & AR hardware comparison

Virtual reality



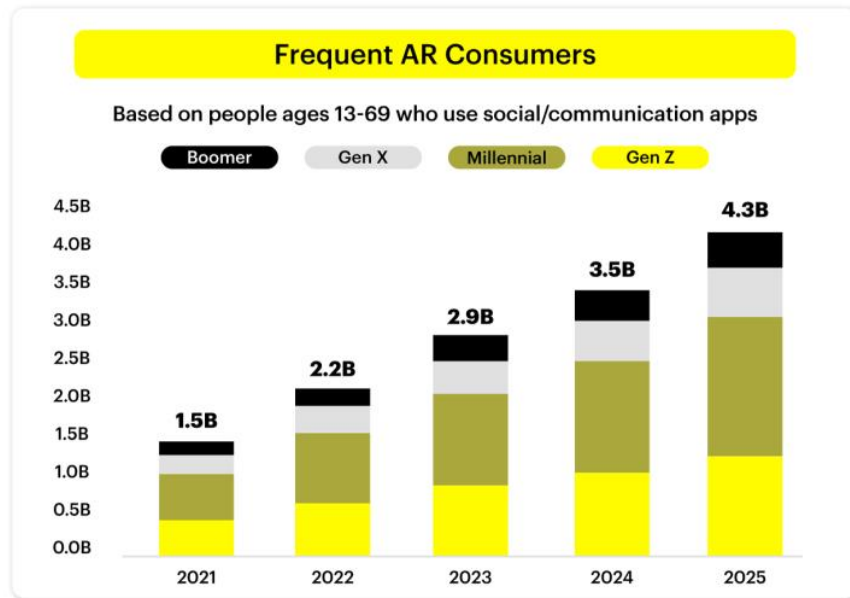
Augmented reality



Future



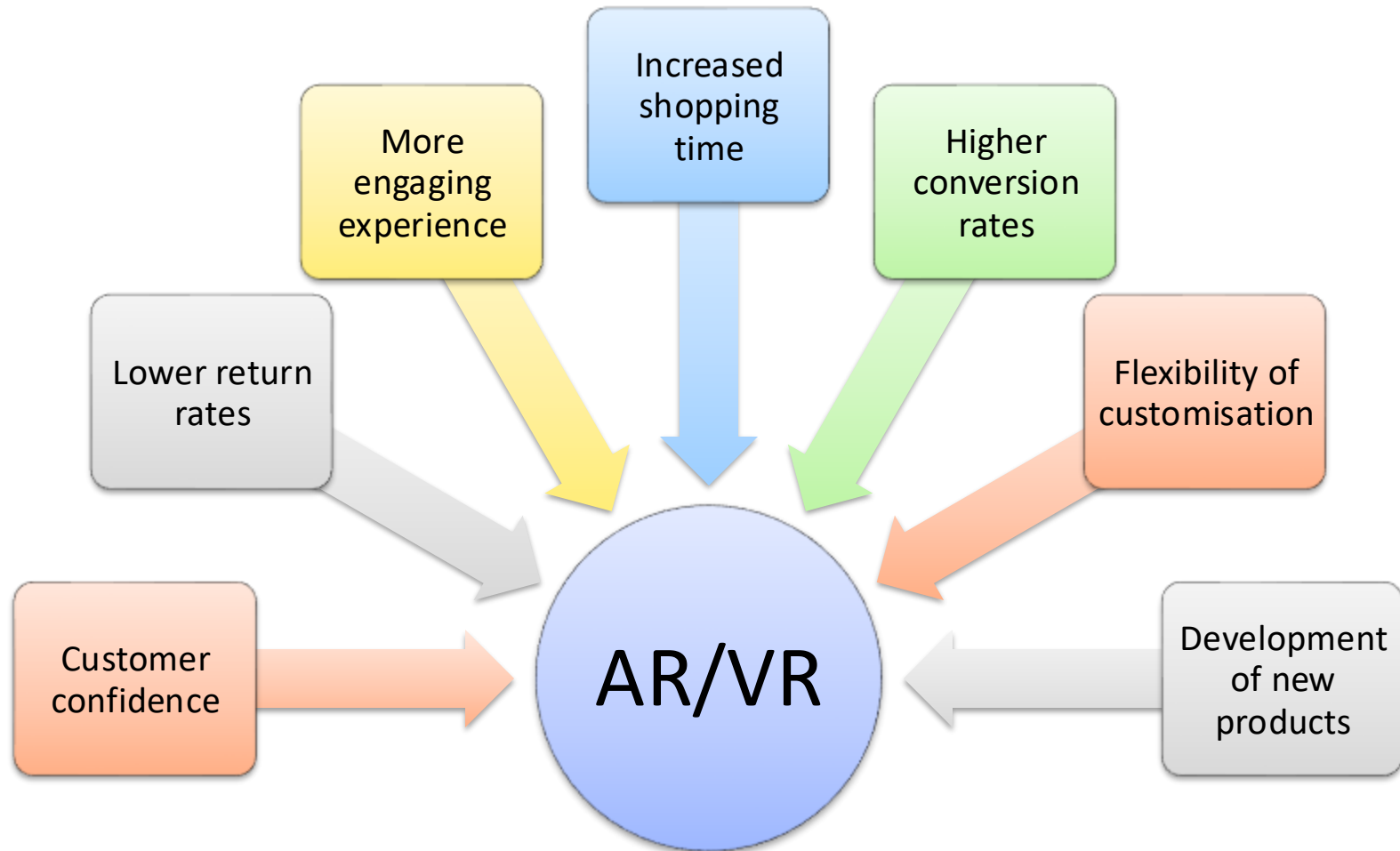
AR/VR in E-commerce #2



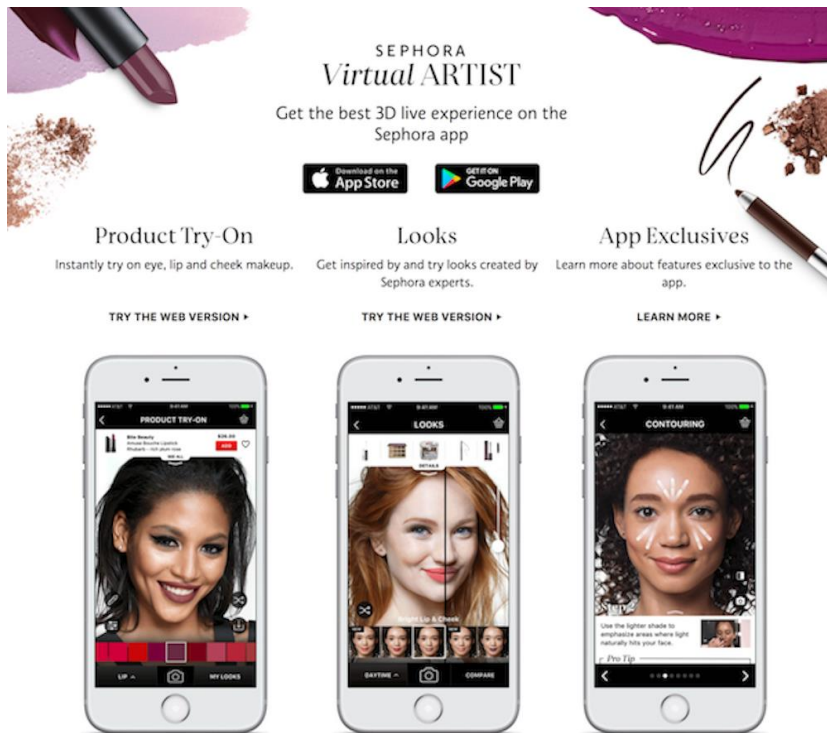
Source: 2021 AR Consumers based on people who use AR weekly or daily from 2021 Global Deloitte Digital Study commissioned by Snap Inc. 2022 to 2025 forecasted in line with 2000-2003 growth rate of smart phone users since AR is in the Toy-phase of adoption; If we look at where this transition took place with mobile phones, one can argue that this was when games like 'snake' started being available (e.g., 1997: Nokia 6110 device released, 1999: emojis were invented, 2000: Nokia 3310 launched, and the first commercially available camera phone launched in Japan).



AR/VR in E-commerce #3



AR/VR in E-commerce #4

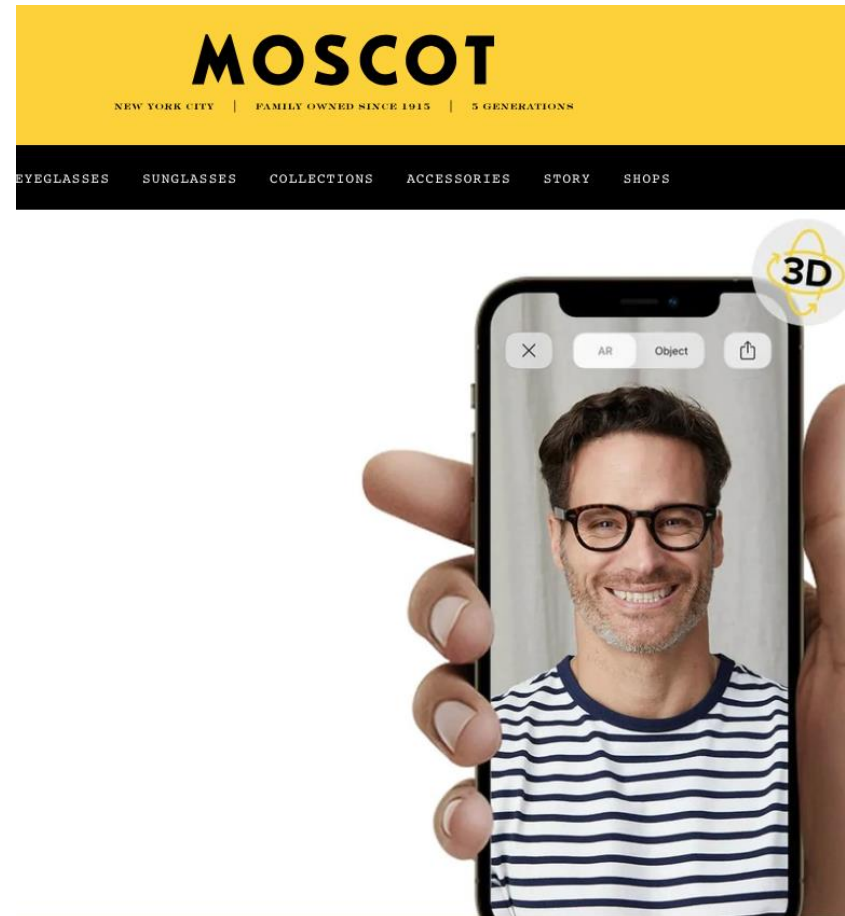


AR/VR in E-commerce #5



Aug

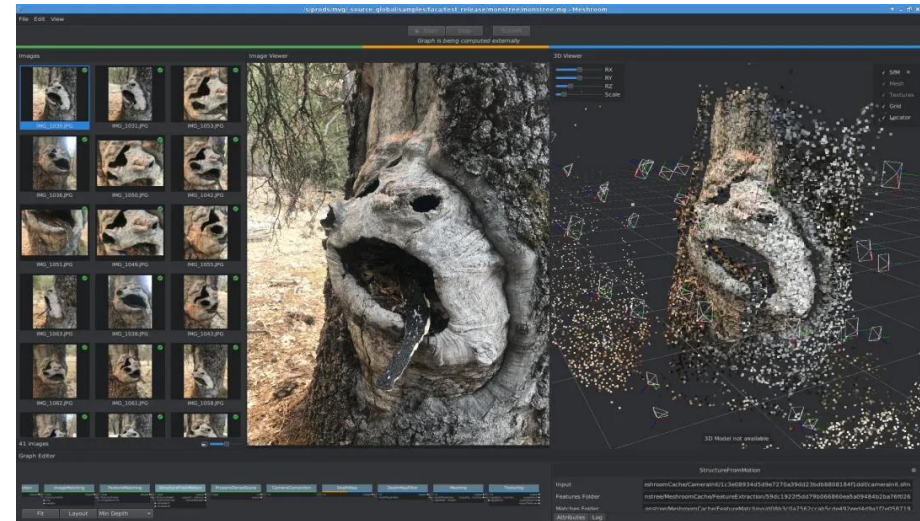
Here you
mobile s
real proc



Constant improvements

3D Scanning - S3D Enscan SE

Photogrammetry - Meshroom



Digital-transformable-to-tangible products

- 3D printing is an emerging market
- We buy 3D printers & printing materials
- How about buying projects to print
- Depends on technology (printers & materials) & economics of printing



ThingMaker – 3D Printer by Mattel (for kids)

COVID-19 & E-commerce – Key points

- The COVID-19 had very significant effect on e-commerce
- It wasn't a temporary situation, the outcome is persistent, creating an opportunity otherwise not achievable (an enforced transition)
- Effects on technology, business and society
- **Virtualisation unfortunately creates some negative social effects ...**

Future of E-commerce



New



Framework Laptop
(12th Gen Intel® Core™)

Starting at €1159

Ready to go with Windows 11

[Product story](#) [Pre-order now](#)

New



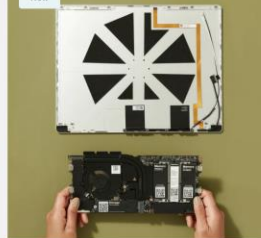
DIY Edition
(12th Gen Intel® Core™)

Starting at €959

Build it yourself and bring your OS

[Product story](#) [Pre-order now](#)

New



Upgrade Kit
(12th Gen Intel® Core™)

Starting at €588

Mainboard & Top Cover (CNC)

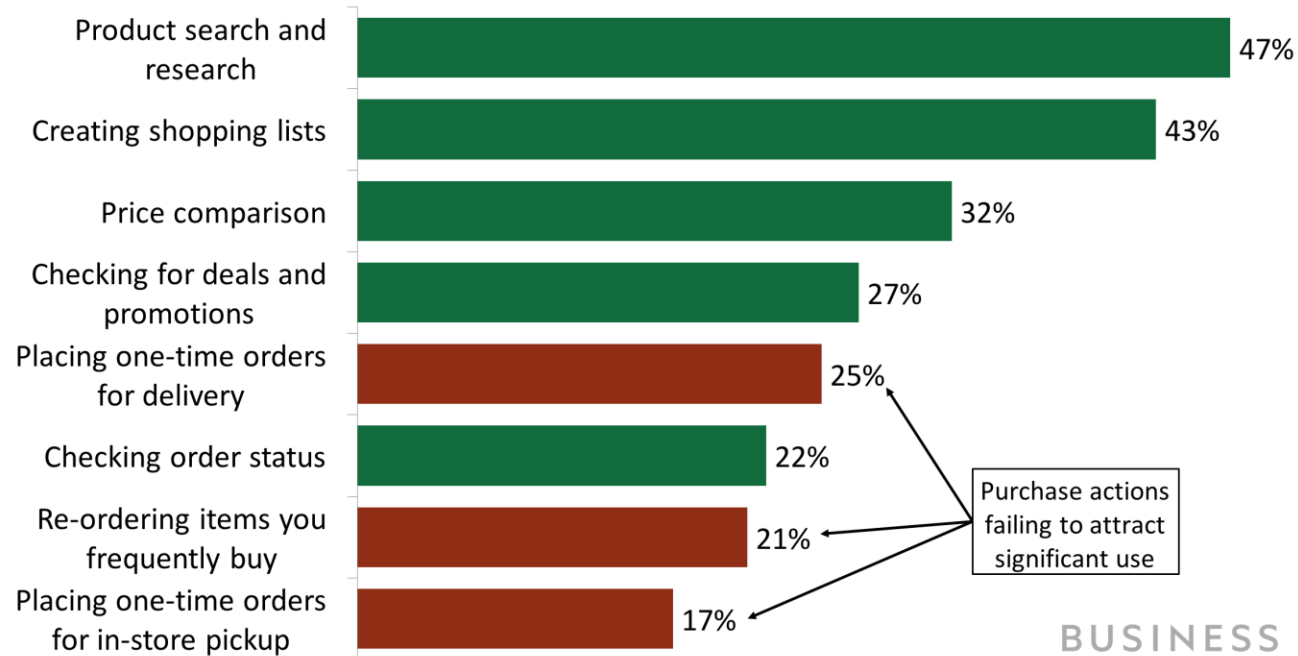
[Get notified](#)



What people need

US Consumers Are Using Smart Speakers For Shopping Prep But Not To Buy Products

How smart speaker owners use voice assistants for shopping

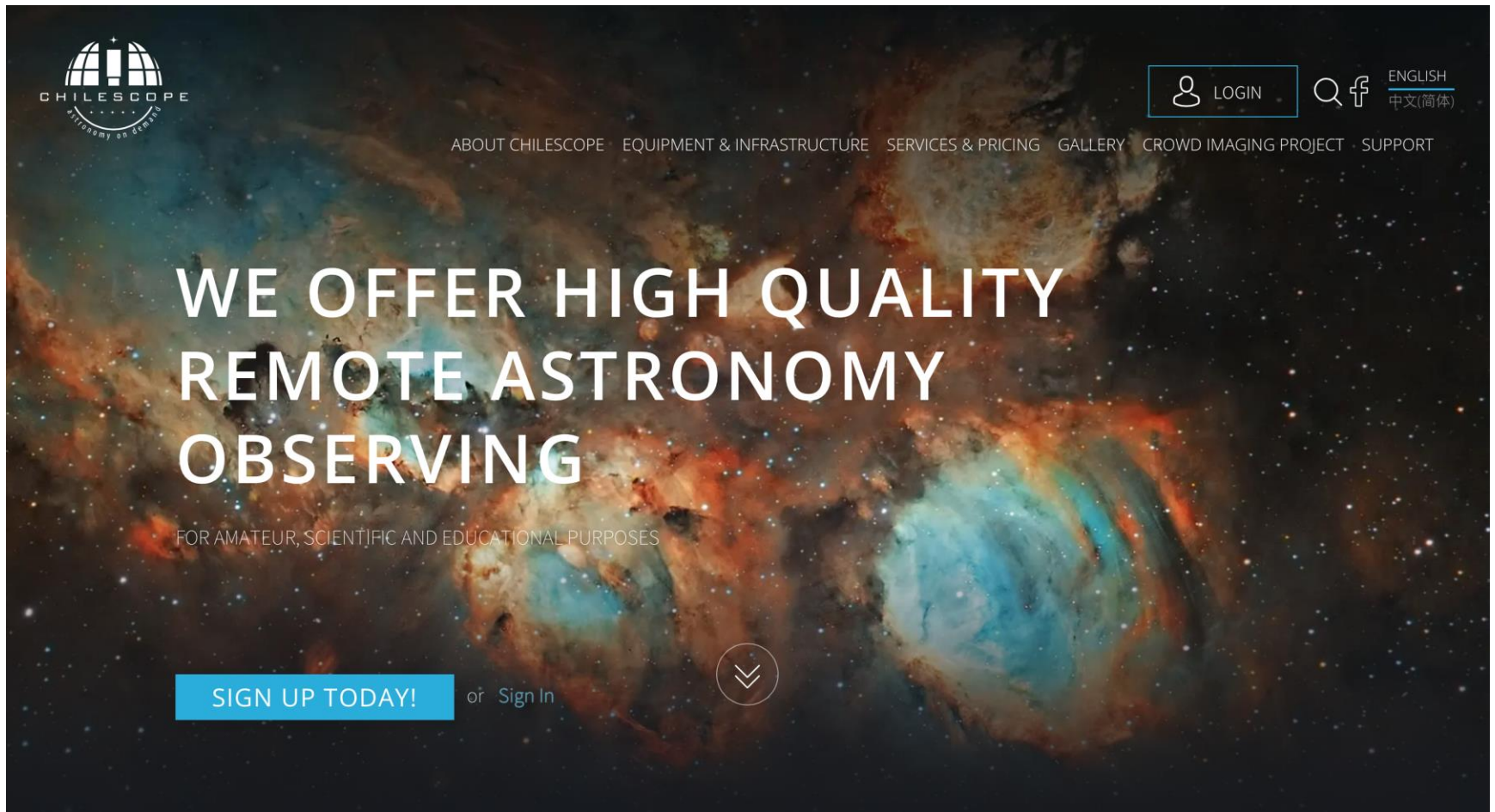


*Note: Values do not total to 100% because respondents could choose multiple answers.
Source: Adobe Digital Insights, n=1,000+ consumers, 2018*

BUSINESS
INSIDER
INTELLIGENCE

Constantly look for opportunities

<http://www.chilescope.com>



Digital items – NFT

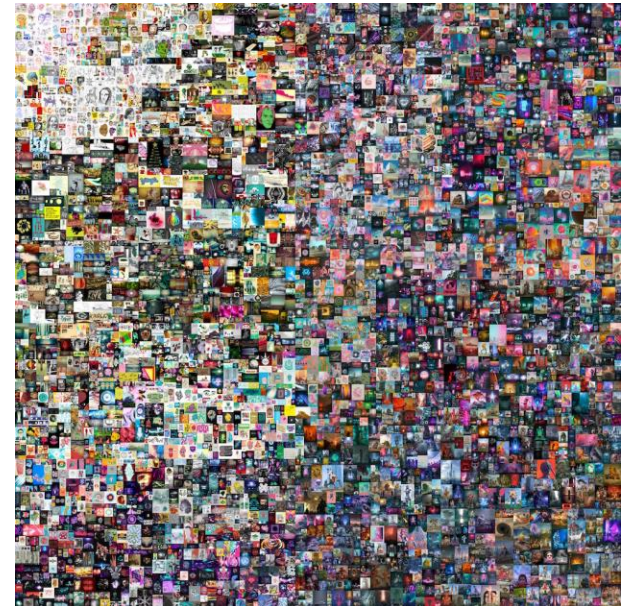
**Kevin Abosch, Forever Rose (2018),
\$1 million, GFTO**



**Paris Hilton and Blake Kathryn, Iconic
Crypto Queen, \$1.111211 million,
Nifty Gateway**



**Beeple, Everydays—The First 5000
Days, \$69 million, Christie's**



Complexity of modern-day EC

ნონა გაფრინდაშვილი - 5 mln \$ case

Nona Gaprindashvili

Queen's gambit



Data-driven E-commerce

Complexity of modern-day OM

“Taiwan is the first country that can watch F9”

John Cena

Fast and Furious 9



PREPARE FOR THE EXAM!